

TIMELESS CAR RENTAL SYSTEM

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TIMELESS CAR RENTAL SYSTEM

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This report is submitted in partial fulfillment of the requirements for the Bachelor of Computer Science (Database Management) with Honours.

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY
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2025

DECLARATION

I hereby declare that this project report entitled

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is written by me and is my own effort and that no part has been plagiarized
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SUPERVISOR :  _____ Date : 24 JUNE 2025
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DEDICATION

This Final Year Project is dedicated with deepest gratitude and appreciation to the many individuals who have supported and inspired me throughout this challenging yet rewarding journey.

I dedicate this project to my loving parents, whose endless support, encouragement, and sacrifices have been the foundation of my journey. Your unwavering belief in me, your patience through every challenge, and your constant motivation have meant more than words can express. This achievement is as much yours as it is mine, and I am forever thankful for everything you have done.

To my lecturers and academic mentors, I sincerely appreciate your dedication to teaching and your guidance throughout my studies. Your insightful advice, constructive feedback, and willingness to share your knowledge have played a crucial role in the success of this project. You have challenged me to grow, think critically, and aim higher, and I am deeply grateful for the impact you've had on my academic development.

To my dear friends and fellow students, thank you for your companionship, support, and the unforgettable memories we've created together. From late-night work sessions to shared laughs during stressful times, your presence has made this journey not only manageable but truly enjoyable. I am grateful for the bonds we've formed and the strength we've shared.

Finally, I dedicate this project to the Almighty Allah SWT, for blessing me with the strength, health, patience, and perseverance needed to complete this Final Year Project. All achievements are only possible through Your guidance and mercy.

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To my friends and fellow classmates, thank you for the support, collaboration, and shared experiences that made this journey more meaningful and enjoyable. Your encouragement and teamwork have helped me stay focused and optimistic throughout this project.

Lastly, I would like to thank myself for not giving up, for pushing through the challenges, and for believing in my abilities even during the toughest times. This journey has taught me resilience, and I am proud of how far I've come.

ABSTRACT

The TimeLess Car Rental System is a web-based platform developed to automate and modernize the traditional manual car rental processes. As digital transformation continues to reshape service industries, the car rental sector faces increasing demand for efficient, user-friendly, and accessible online systems. Currently, many car rental businesses rely on manual communication methods such as WhatsApp or Telegram, leading to booking delays, data mismanagement, and overbooking issues. This project addresses these challenges by designing a system that allows customers to browse available vehicles, make reservations, and perform transactions online while enabling administrators to manage bookings, monitor vehicle availability, and store customer records efficiently. The system aims to improve booking accuracy, minimize manual errors, and enhance user experience through real-time updates and automation. Using HTML, CSS, JavaScript for the frontend and PHP with MySQL for the backend, the system includes various modules such as customer registration and login, car management, booking and payment processing, and password recovery. The research process involved system analysis, database design, interface development, and testing to ensure functionality and user satisfaction. Results indicate that the TimeLess Car Rental System successfully reduces administrative burden, prevents double bookings, and increases customer convenience by enabling 24/7 access to services. This project ultimately demonstrates how digital platforms can optimize operational workflows and improve service quality in the car rental industry.

ABSTRAK

Sistem TimeLess Car Rental merupakan sebuah platform berasaskan web yang dibangunkan untuk mengautomasikan dan memodenkan proses penyewaan kereta secara manual. Seiring dengan arus transformasi digital yang pesat dalam pelbagai industri perkhidmatan, sektor penyewaan kereta turut berhadapan dengan permintaan yang semakin meningkat terhadap sistem dalam talian yang cekap, mesra pengguna dan mudah diakses. Pada masa ini, kebanyakan perniagaan penyewaan kereta masih bergantung kepada kaedah komunikasi manual seperti WhatsApp atau Telegram, yang boleh menyebabkan kelewatan dalam pengesahan tempahan, pengurusan data yang tidak sistematik, serta isu tempahan berganda. Projek ini dibangunkan bagi menangani masalah tersebut dengan membina satu sistem yang membolehkan pelanggan menyemak ketersediaan kenderaan, membuat tempahan, serta melaksanakan pembayaran secara atas talian. Selain itu, pentadbir juga dapat mengurus tempahan, memantau ketersediaan kenderaan, dan menyimpan maklumat pelanggan secara lebih tersusun. Sistem ini direka untuk meningkatkan ketepatan tempahan, mengurangkan ralat manual, serta mempertingkatkan pengalaman pengguna menerusi kemas kini masa nyata dan automasi proses. Pembangunan sistem melibatkan penggunaan HTML, CSS, JavaScript bagi antara muka hadapan serta PHP dan MySQL bagi pengaturcaraan belakang. Modul-modul utama termasuklah pendaftaran dan log masuk pelanggan, pengurusan kereta, tempahan dan pembayaran, serta pemulihan kata laluan. Proses kajian meliputi analisis sistem, reka bentuk pangkalan data, pembangunan antara muka pengguna dan ujian fungsi sistem. Hasil kajian menunjukkan bahawa sistem ini berjaya mengurangkan beban pentadbiran, mengelakkan tempahan berganda, serta meningkatkan kemudahan kepada pelanggan dengan membolehkan akses 24 jam kepada perkhidmatan penyewaan. Secara keseluruhan, projek ini membuktikan bahawa platform digital dapat mengoptimumkan aliran kerja operasi dan meningkatkan kualiti perkhidmatan dalam industri penyewaan kereta.

TABLE OF CONTENTS

	PAGE
DECLARATION.....	II
DEDICATION.....	III
ACKNOWLEDGEMENTS.....	IV
ABSTRACT	V
ABSTRAK	VI
TABLE OF CONTENTS.....	VII
LIST OF TABLES	X
LIST OF FIGURES	XI
LIST OF ABBREVIATIONS	XIII
LIST OF ATTACHMENTS.....	XIV
Chapter 1: INTRODUCTION	1
1.1 Introduction.....	1
1.2 Problem Statement(s).....	2
1.3 Objectives	3
1.4 Scope.....	4
1.5 Project Significance	5
1.6 Expected Output.....	6
1.7 Conclusion	6
CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY .	8
2.1 Introduction.....	8

2.2	Literature Review.....	8
2.2.1	Comparison between existing system and proposed system	8
2.3	Project Methodology.....	11
2.3.1	Database Initial Study.....	12
2.3.2	Database Design	13
2.3.3	Implementation and Loading	14
2.3.4	Testing and Evaluation	14
2.3.5	Operation	15
2.3.6	Maintenance and Evolution	15
2.4	Conclusion	16
CHAPTER 3: ANALYSIS.....		17
3.1	Introduction.....	17
3.2	Problem Analysis	17
3.3	The Proposed Improvement/Solution	18
3.4	Requirement analysis of the to-be system	20
3.4.1	Functional Requirement (Process Model)	20
3.4.1.1	Data Flow Diagram (DFD)	22
3.4.1.2	Data Flow Diagram (DFD) Level 0.....	23
3.4.2	Non-Functional Requirement	25
3.4.2.1	Quality Requirements	25
3.4.2.2	Performance Requirements.....	27
3.4.3	Other Requirement.....	28
3.4.3.1	Software Requirements.....	28
3.4.3.2	Hardware Requirements	30

3.5	Conclusion	31
CHAPTER 4: DESIGN		32
4.1	Introduction.....	32
4.2	System Architecture.....	32
4.3	Database Design.....	33
4.3.1	Conceptual Design.....	33
4.3.1.1	Entity Relationship Diagram (ERD).....	33
4.3.1.2	Business Rules	33
4.3.2	Logical Design.....	34
4.3.2.1	Data Dictionary for Entity Relationship Diagram	34
4.3.2.2	Query Design	46
4.4	Graphical User Interface (GUI) Design.....	48
4.4.1	Customer Interface.....	48
4.4.2	Admin Interface	56
4.4.3	Delivery Staff Interface	64
REFERENCES.....		66

LIST OF TABLES

	PAGE
Table 2.1 Comparison between existing system and proposed system	8
Table 3.1 Software Requirements.....	28
Table 3.2 Hardware Requirements	30
Table 4.1 Admin Table	34
Table 4.2 Delivery Staff Table	35
Table 4.3 Customer Table	36
Table 4.4 Driver Table.....	37
Table 4.5 Booking Table	38
Table 4.6 Booking Image Table	40
Table 4.7 Car Table	41
Table 4.8 Car Image Table.....	43
Table 4.9 Agreement Form Table.....	43
Table 4.10 Guarantor Table.....	44
Table 4.11 Payment Table	45
Table 4.12 Service Table.....	45
Table 4.13 SQL Table	47

LIST OF FIGURES

	PAGE
Figure 2.1 Database Life Cycle	12
Figure 2.2 Wahdah Homepage	12
Figure 2.3 Avis Homepage.....	13
Figure 3.1 Flowchart of current system	17
Figure 3.2 Flowchart of Proposed System	19
Figure 3.3 Context Diagram.....	22
Figure 3.4 Data Flow Diagram (DFD) Level 0	23
Figure 4.1 ERD.....	33
Figure 4.2 Customer Login Page	48
Figure 4.3 Customer Dashboard Page	49
Figure 4.4 Book Car Page.....	50
Figure 4.5 Driver Details Page	51
Figure 4.6 Guarantor's Details Page	51
Figure 4.7 Review Booking Page	52
Figure 4.8 Review Booking Page	52
Figure 4.9 Rental Agreement and Signature Page.....	53
Figure 4.10 Booking Submission Page	54
Figure 4.11 Customer Booking Page	55
Figure 4.12 Payment Page	55
Figure 4.13 Admin Login Page	56
Figure 4.14 Admin Dashboard Page	56
Figure 4.15 Admin Approval Booking Page.....	57
Figure 4.16 Inspection Pickup and Return Page.....	58
Figure 4.17 Admin Manage Customer Page.....	58

Figure 4.18 Manage Cars Page	59
Figure 4.19 Manage Bookings Page.....	60
Figure 4.20 Drivers and Guarantor Details Page.....	60
Figure 4.21 Payment and Refunds Page	61
Figure 4.22 Add Delivery Staff Page	61
Figure 4.23 Services and Delivery Assignment	62
Figure 4.24 Monthly Income Report Page.....	63
Figure 4.25 Most Popular Cars Report Page.....	64
Figure 4.26 Delivery Staff Login.....	64
Figure 4.27 Delivery Staff Dashboard Page.....	65

LIST OF ABBREVIATIONS

FYP - **Final Year Project**

LIST OF ATTACHMENTS

	PAGE
Appendix A	
Sample of data	19
Appendix B	
Analysis of data collection	78
.....	
.....	

Chapter 1: INTRODUCTION

1.1 Introduction

In recent years, the car rental industry has experienced significant growth, driven by rising customer demand for flexible, efficient, and technology-driven services. Despite this shift, many car rental businesses continue to rely on manual systems for booking, communication, and recordkeeping. This outdated approach often leads to inefficiencies such as delays in confirming reservations, miscommunication between customers and car owners, overbooking, and difficulty in managing vehicle and customer data.

The TimeLess Car Rental System is proposed as a digital solution to overcome these challenges. It is a web-based system designed to replace the traditional manual methods with an automated, centralized platform. This system enables customers to conveniently browse available vehicles, make bookings, and receive instant confirmations online. Meanwhile, business owners (admins) can efficiently manage car inventories, track bookings in real time, and store records securely without the need for paperwork.

This project not only aims to improve the customer experience by offering an accessible and user-friendly interface but also enhances operational efficiency for car rental service providers. With automation, the risk of human error in bookings is reduced, and the overall workflow is streamlined. Additionally, the system includes features such as login and registration modules, password recovery, payment integration, and a dashboard for administrators to oversee and manage their operations.

In summary, the TimeLess Car Rental System seeks to modernize car rental management by leveraging web technologies to create a reliable, efficient, and scalable platform that benefits both customers and business owners.

1.2 Problem Statement(s)

While car rental services continue to be in high demand, especially in urban areas, many rental businesses still rely on outdated manual processes, which result in numerous operational challenges:

- **Manual booking and communication methods:** Most car rental businesses still use messaging apps like WhatsApp or Telegram to handle bookings and customer inquiries. This leads to slow response times, miscommunication, and inconsistent service delivery. Customers often face delays in receiving confirmation, which can result in dissatisfaction and booking conflicts.
- **Paper-based recordkeeping:** Business owners frequently manage vehicle details, customer information, and booking histories manually using notebooks or spreadsheets. This increases the risk of data loss, duplication, and human error. As the business grows, managing records without a centralized system becomes inefficient and time-consuming.
- **Limited accessibility for customers:** Without a dedicated online platform, customers cannot easily browse available vehicles, check booking availability, or make reservations independently. This lack of self-service options restricts convenience and can drive potential customers away.
- **Lack of real-time tracking and automation:** Traditional systems offer no real-time car availability tracking or automated booking confirmation. This results in overbooking issues, scheduling conflicts, and missed opportunities for both the customer and the business owner.

These problems highlight the need for a comprehensive web-based solution that can digitalize the car rental process, enhance communication, and improve operational efficiency through automation and centralized data management.

1.3 Objectives

This project is developed with the aim of addressing the inefficiencies of the current manual car rental process. The main objectives are as follows:

- **Digitalize the car rental process:** To replace traditional booking methods such as WhatsApp or Telegram with an online web-based system. This allows customers to view available cars, make bookings, and receive confirmations instantly without needing to wait for manual replies.
- **Centralize data management:** To create a platform where all vehicle, booking, and customer information is stored in one place. This reduces dependency on paper records and spreadsheets, ensuring easier data access, better organization, and fewer human errors.
- **Enable real-time car availability tracking:** To allow both customers and admins to see up-to-date car availability, which helps avoid overbooking, improves scheduling accuracy, and enhances the user experience.
- **Improve business efficiency and decision-making:** To provide an admin dashboard that offers clear visibility of bookings, payment records, and fleet details, enabling business owners to manage operations more effectively and make data-driven decisions.
- **Enhance customer convenience and accessibility:** To allow customers to book rentals anytime and anywhere through a user-friendly interface, thus improving satisfaction and expanding the service's reach.
- **Automate administrative tasks:** To reduce the workload of car rental owners by automating booking confirmations, payment tracking, and customer record handling, thereby increasing productivity and reducing operational errors.

1.4 Scope

- **User Demographic:** The primary users of the TimeLess Car Rental System are customers, admins (car rental business owners), and delivery staff. Customers are individuals who register an account on the system, browse available cars, and make bookings by entering driver and guarantor details. After booking, they must wait for admin verification before proceeding to payment. Admins manage all aspects of the system, including car inventory, customer information, bookings, payments, refunds, and delivery staff assignments. Delivery staff are appointed by admins to handle the physical delivery of vehicles to customers.
- **Geographical Scope:** The system will initially cater to users within a specific city or region to ensure efficient delivery logistics and localized service management. However, it is designed to be scalable and may later be expanded to serve multiple branches or operate at a national level.
- **Services Offered:** The system offers several essential services, including online browsing and booking of cars, submission of driver and guarantor information, admin booking verification before payment, and acceptance of a digital agreement form. Customers can proceed with payments via online banking or credit card by selecting a preferred bank and entering their card number, CVV, and expiration date. Refunds are handled by admins when necessary. Admins also assign delivery staff for approved bookings and have access to a secure dashboard to manage all operations. These features aim to eliminate the delays, errors, and communication issues associated with manual car rental processes.
- **Booking and Rental Management:** The platform enables efficient handling of booking requests, tracks vehicle availability in real-time, and helps prevent overbooking. Booking terms are enforced through an integrated digital agreement that customers must accept. Admins have full control over each booking's status and history, ensuring smooth transactions and reliable customer service.

- **Reporting and Analytics:** Admins can generate and view income reports on a daily and monthly basis through a built-in reporting module. These reports provide insights into business performance and assist in financial management and decision-making.

1.5 Project Significance

The significance of this project lies in its ability to revolutionize the car rental process by offering a centralized, automated, and user-friendly digital platform that enhances service efficiency, minimizes errors, and improves overall user

- **Customers:** Customers benefit from a streamlined and user-friendly process for renting vehicles. They can register, view available cars, submit bookings with detailed driver and guarantor information, and proceed with secure payments via online banking or credit card. The requirement for admin booking verification ensures order accuracy and prevents misuse. The system eliminates manual communication, provides instant status updates, and builds customer confidence through a faster, more transparent rental experience.
- **Admins (Car Rental Business Owners):** Admins gain complete control over the car rental operation through a centralized dashboard. They can manage customer accounts, car inventory, bookings, drivers, guarantors, and delivery assignments. The system also allows admins to verify bookings before payment, manage refunds, and monitor business performance through daily and monthly income reports. This digital transformation significantly reduces paperwork, enhances accuracy, and supports data-driven decision-making.
- **Delivery Staff:** Delivery staff are given individual login access and a dedicated dashboard where they can view cars assigned for delivery by the admin. They are responsible for updating the delivery status of each vehicle, ensuring real-time tracking and accountability. This feature improves coordination between the admin and delivery team, helping ensure timely and reliable customer service.

1.6 Expected Output

The expected outputs from this project are as follows:

- **Output 1:** A comprehensive customer dashboard where users can register, browse available cars, submit bookings with driver and guarantor information, and proceed with secure payments via online banking or credit card. The dashboard will provide real-time updates on booking status and vehicle availability, enhancing user experience and transparency.
- **Output 2:** A centralized admin portal for car rental business owners to manage the entire rental process. This includes handling customer data, car listings, booking verifications, payment and refund processing, delivery staff assignments, and generating daily and monthly income reports. The portal also allows for monitoring of business performance and ensures accurate, paperless operations.
- **Output 3:** A dedicated delivery staff dashboard where delivery personnel can log in to view assigned deliveries, update delivery statuses, and ensure timely handover of vehicles to customers. This feature enables better coordination and real-time tracking of the delivery process, improving customer satisfaction and operational reliability.
- **Output 4:** An integrated agreement module where customers are required to review and accept the terms and conditions before finalizing any booking. This module ensures compliance, reduces disputes, and builds trust between the customer and the rental service.

1.7 Conclusion

In conclusion, this chapter has presented the background, objectives, problem statement, scope, significance, and expected outputs of the TimeLess Car Rental System project. The project aims to develop a comprehensive car rental management system that digitizes the booking process, enhances coordination between customers, admins, and delivery staff, and improves overall efficiency and service quality. By

automating manual processes and integrating key features such as booking verification, secure online payments, and delivery tracking, the system is expected to deliver a seamless and reliable rental experience. The next chapter will focus on the literature review, examining existing systems, related technologies, and previous research to inform the design and development of the proposed solution.

CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Introduction

This chapter presents a comprehensive overview of relevant literature and the methodology adopted for the development of the TimeLess Car Rental System. The literature review explores existing car rental systems, related technologies, frameworks, and best practices that inform the project's design and implementation. By analyzing similar systems and identifying their strengths and limitations, this review provides valuable insights to ensure the proposed system meets modern user expectations and business needs. The second part of the chapter outlines the project methodology, detailing the systematic approach taken to analyze requirements, design system components, implement features, and test the functionalities. This structured approach ensures the project is developed efficiently, aligns with user needs, and delivers the intended outcomes.

2.2 Literature Review

2.2.1 Comparison between existing system and proposed system

Table 2.1 Comparison between existing system and proposed system

Feature	Wahdah	Avis	Proposed System
User Registration and Login	Web-based registration & login for customers. Admin login via Driveo portal.	Users can book without logging in; account creation only for managing reservations.	Web-based registration and login for customers, admin (car rental owner) & delivery staff. Forgot password included.

Real-Time Availability	Customers can view real-time availability of cars based on location & date.	Car availability shown as static listings; requires manual inquiry	Real-time availability tracking by admin and visible to customers. Prevents overbooking.
Booking Process	Online booking via calendar-based system with pickup/drop-off selection.	Form-based booking; requires user to submit a request for reservation confirmation.	Booking via web form with driver and guarantor details. Admin must verify booking before payment.
Payment Integration	Supports credit card, FPX, and e-wallet payments during checkout.	Credit card payment integrated; payment due at booking or pickup.	Online banking & credit card payments (card number, CVV, expiry). Payment only after admin approval.
Reporting and Analytics	Available via Driveo admin portal for partners/car owners to view earnings and performance.	Not accessible to users; analytics likely internal for corporate use only.	Admin can view daily and monthly income reports for financial tracking and decision-making.

Admin Control and Maintenance	Admin/partner portal (Driveo) allows car owners to manage cars, bookings, documents, and earnings.	Centralized admin portal not public; corporate-level admin control	Full admin dashboard to manage cars, bookings, users, payments, refunds, delivery, and system reports.
Security Features	HTTPS secured, OTP/email verification, encrypted transactions.	Standard HTTPS, data protection, and secure booking	Secure login, booking verification, agreement enforcement, and restricted role-based access (e.g. delivery staff).

Based on Table 2.1, Wahdah Malaysia offers a fairly comprehensive digital car rental experience with features like user registration, real-time car availability, integrated payment, and an admin portal that allows car owners to manage bookings and access reporting tools. This makes Wahdah a reliable platform for individual rental businesses operating under a structured system. However, it still has some limitations in customization for independent business processes.

Avis Car Rental, on the other hand, provides a more traditional and corporate-centric service. While it supports online booking and credit card payments, its system lacks real-time availability tracking and admin-level access or reporting tools. The process is more static, requiring manual confirmation and offering less flexibility, especially for small or independent rental operators.

In contrast, the proposed TimeLess Car Rental System is designed to bridge the gaps identified in both existing systems. It supports full user registration, real-time

availability, structured booking (including driver and guarantor info), secure online payment via banking or credit card, and most importantly, a powerful admin dashboard. This dashboard includes features such as booking verification, refund handling, delivery staff assignment, and income reporting. Moreover, it offers a guarantor details, a unique feature not found in Wahdah or Avis. Overall, the TimeLess system is better aligned with the needs of small-to-medium rental businesses seeking complete digital control, efficiency, and scalability.

2.3 Project Methodology

The development of the TimeLess Car Rental System followed a structured method based on both the Software Development Life Cycle (SDLC) and the Database Life Cycle (DBLC). The project started with gathering requirements from the admin (who is also the car rental owner), customers, and delivery staff to understand their needs. This included features like booking verification, driver and guarantor details, payment handling, delivery updates, and income reporting. The system design was done using MySQL in phpMyAdmin, and the interfaces were designed to be simple and user-friendly for each type of user. The database was normalized to ensure data accuracy and better performance.

The system was developed using PHP for backend and HTML, CSS, and JavaScript for the frontend. It was hosted on the Apache web server with a MySQL database. Testing was done through unit testing, integration testing, and user acceptance testing to make sure the system works correctly and meets expectations. The delivery staff dashboard, admin reports, and booking flow were all tested. This system is suitable for small car rental businesses where the admin is also the owner, helping them manage their daily operations more easily in one complete system.

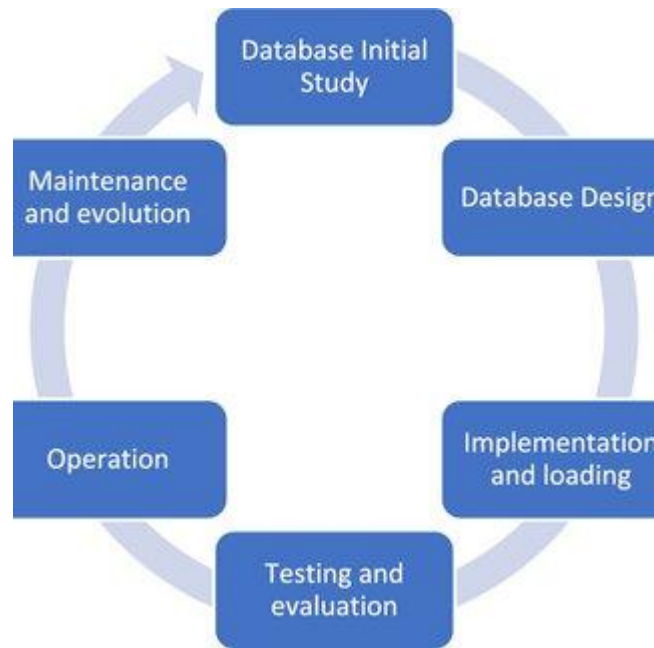


Figure 2.1 Database Life Cycle

2.3.1 Database Initial Study

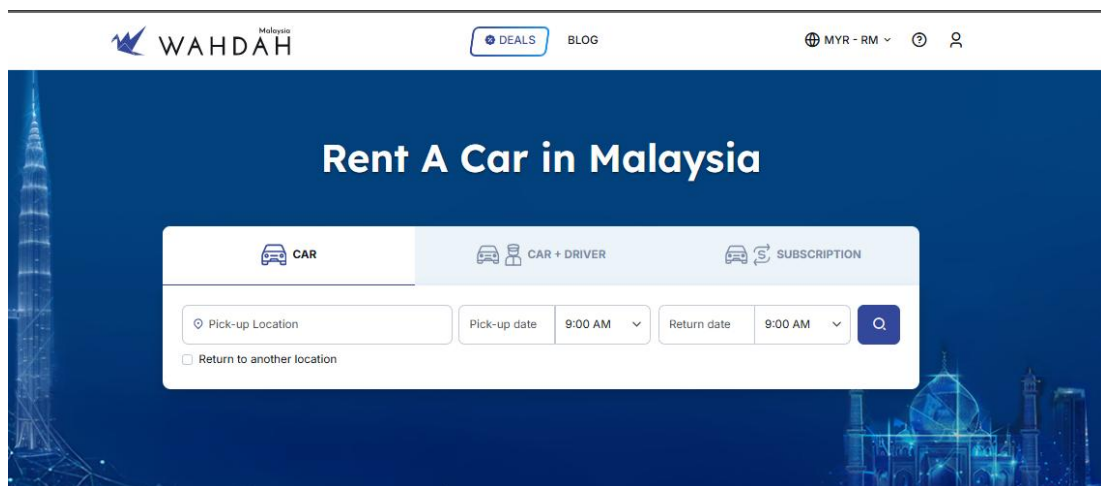


Figure 2.2 Wahdah Homepage

Wahdah Malaysia is an online car rental platform that allows users to rent vehicles easily across various locations in Malaysia. The homepage offers a clean, user-friendly interface where customers can search for cars by selecting pick-up and return locations, dates, and times. Users can choose between renting just a car, a car with a driver, or subscribing to rental packages. The platform supports flexible return options

and provides real-time availability, ensuring a convenient and smooth booking process. Wahdah aims to simplify car rentals through a digital-first approach that caters to both local and international users.

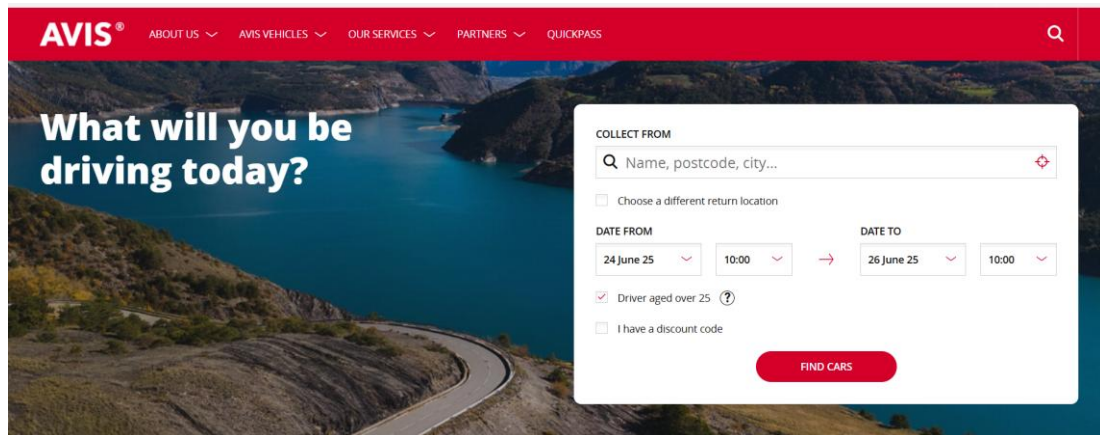


Figure 2.3 Avis Homepage

Avis Malaysia is a global car rental service provider that operates in major cities across Malaysia, including Melaka. The platform offers a professional and corporate-style online booking system where users can select their rental location, pick-up and return dates, and vehicle preferences. Avis supports services for both leisure and business travelers, with options ranging from economy to luxury vehicles. While the interface is straightforward, booking confirmation often requires follow-up, and vehicle availability may not be shown in real time. Avis focuses on reliability, brand reputation, and customer service, making it ideal for users seeking established, premium rental services.

2.3.2 Database Design

The second phase of this project focuses on the design of the database model to support the operational needs and features of the TimeLess Car Rental System. This phase includes three main sub-phases: conceptual design, logical design, and physical design. In the conceptual design stage, an Entity-Relationship Diagram (ERD) was created to represent the main entities such as Admin, Customer, Car, Booking, Payment, Driver, Guarantor, and Delivery Staff. The ERD was normalized to ensure consistency and to avoid data redundancy. During the logical design, relationships between entities were refined and attributes were defined based on system

requirements such as booking verification, payment tracking, and delivery status updates. In the physical design phase, the database was implemented in MySQL using phpMyAdmin, where tables, columns, primary and foreign keys, constraints, and indexes were created to reflect the business rules and support smooth system performance. This design ensures accurate data handling and supports the system's overall functionality and reliability.

2.3.3 Implementation and Loading

The implementation and loading phase of this project involves the actual development and deployment of the TimeLess Car Rental System. This phase focuses on transforming the system design and database model into a working application. Key modules such as customer and delivery staff registration and login, car listing management, booking with driver and guarantor details, admin dashboard, booking verification, payment integration, refund handling, delivery assignment, and income reporting are developed based on the requirements from earlier phases. This stage includes writing and integrating code using PHP, HTML, CSS, JavaScript, and MySQL, along with testing each component for proper functionality. Debugging and feature integration are carried out to ensure that all modules work smoothly together. Additionally, sample data is inserted into the database to support initial testing and simulate real-world scenarios. Quality assurance activities such as unit testing, integration testing, and user acceptance testing are conducted to confirm that the system performs reliably and meets user expectations. Once verified, the system is made ready for operational use and future deployment.

2.3.4 Testing and Evaluation

The testing and evaluation phase is crucial in ensuring the reliability, functionality, and performance of the TimeLess Car Rental System. Various testing methods such as unit testing, integration testing, system testing, and user acceptance testing are carried out to detect and fix errors or bugs. This phase ensures that all features including customer and delivery staff login, car availability display, booking with driver and guarantor information, admin verification, payment processing, refund handling, and report generation work as expected and meet the defined system requirements. Feedback from users is collected and reviewed to further improve the system's

usability and efficiency. Evaluation is based on key aspects such as user experience, response time, accuracy of data handling, system scalability, and security. The goal is to verify that the system is fully functional, user-friendly, and ready for deployment in real-world rental operations.

2.3.5 Operation

The operation phase marks the official launch and live use of the TimeLess Car Rental System. After completing all testing and evaluation activities, the system is deployed for real-world use by the admin (car rental owner), customers, and delivery staff. During this phase, users begin interacting with the system to perform key tasks such as registering accounts, browsing available cars, making bookings, verifying driver and guarantor details, processing payments, and updating delivery status. Admins use the system to manage bookings, assign delivery staff, approve payments, and generate income reports. Continuous system monitoring and support are provided to ensure smooth performance and quick resolution of any operational issues. This phase focuses on maintaining system stability, ensuring data accuracy, and providing a seamless user experience to support the day-to-day operations of the car rental business.

2.3.6 Maintenance and Evolution

Maintenance and evolution focus on supporting and improving the TimeLess Car Rental System after it has been deployed. Regular maintenance tasks such as system updates, bug fixes, and security patches are carried out to ensure the system remains stable, secure, and efficient. User feedback from admins, customers, and delivery staff is continuously reviewed to identify any issues or areas for improvement. As the business grows or user needs change, the system evolves through planned feature enhancements, such as adding new reporting options, refining the booking flow, or integrating new payment methods. This phase requires ongoing collaboration between the developer and the car rental owner (admin) to prioritize updates while minimizing disruption to daily operations. Continuous evaluation and improvements help extend the system's lifespan, enhance user satisfaction, and ensure the TimeLess Car Rental System remains relevant and effective for long-term business use.

2.4 Conclusion

In conclusion, the development of the TimeLess Car Rental System has followed a structured approach based on the principles of the Database Life Cycle (DBLC), ensuring systematic phases of planning, design, implementation, testing, deployment, and maintenance. The project successfully addressed key operational needs such as booking management, payment processing, driver and guarantor details, admin verification, delivery coordination, and income reporting. Using MySQL with phpMyAdmin for database design and PHP with HTML, CSS, and JavaScript for backend and frontend development, the system was designed to be both functional and user-friendly. Comprehensive testing methods, including unit, integration, and user acceptance testing, were applied to ensure the system's reliability, accuracy, and security. The deployment of the system allows admins (car rental owners) to efficiently manage bookings and operations, while customers and delivery staff benefit from simplified processes and real-time updates. Ongoing maintenance and future improvements will allow the system to evolve according to user needs and business growth, making it a practical and scalable solution for small car rental businesses seeking digital transformation and enhanced service delivery.

CHAPTER 3: ANALYSIS

3.1 Introduction

This chapter delves into the analysis phase of the project, providing a detailed examination of the current system, proposed improvements, and requirements of the to be system. The analysis will be developed through a systematic approach, ensuring a thorough understanding of the system's needs and potential solutions.

3.2 Problem Analysis

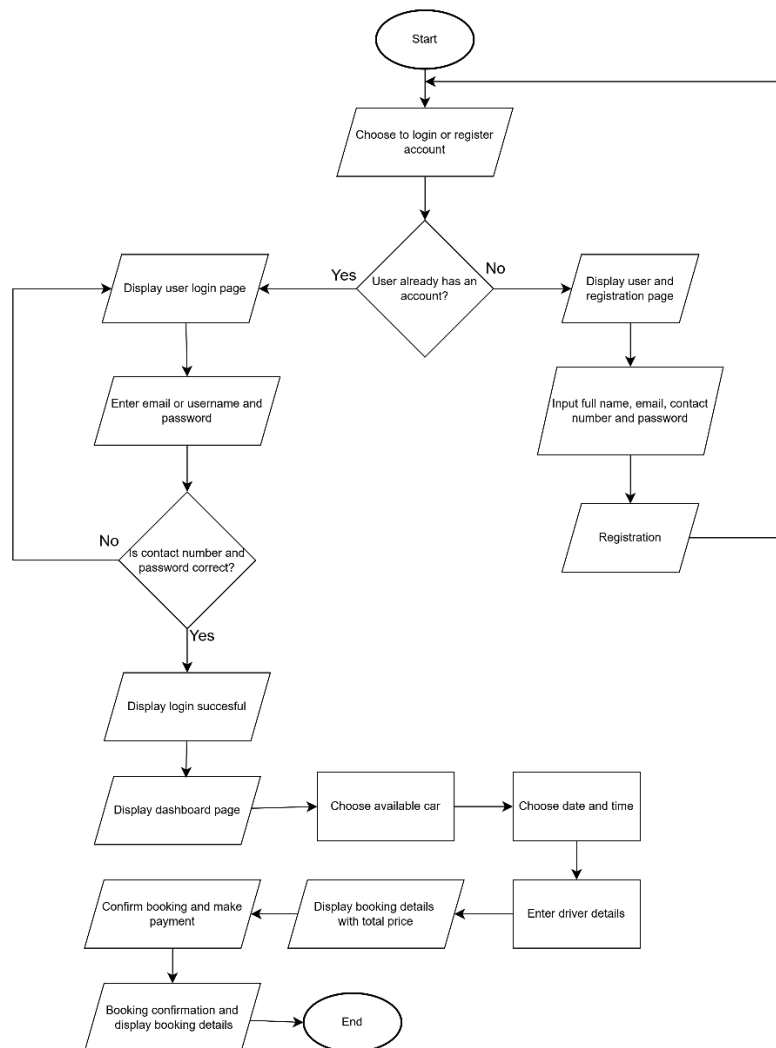


Figure 3.1 Flowchart of current system

Based on Figure 3.1, the process begins when a customer wants to rent a car. The customer visits the car rental website and searches for available vehicles by selecting the pickup location, return location, and rental dates. The system displays a list of available cars that match the search criteria. The customer then selects a preferred car and proceeds to fill in the required driver details. Once the information is entered, the customer is directed to make a payment online using a debit or credit card or other available methods. After the payment is completed, the customer receives an instant confirmation from the system. The booking details are usually followed up manually via WhatsApp or email by the car rental provider.

3.3 The Proposed Improvement/Solution

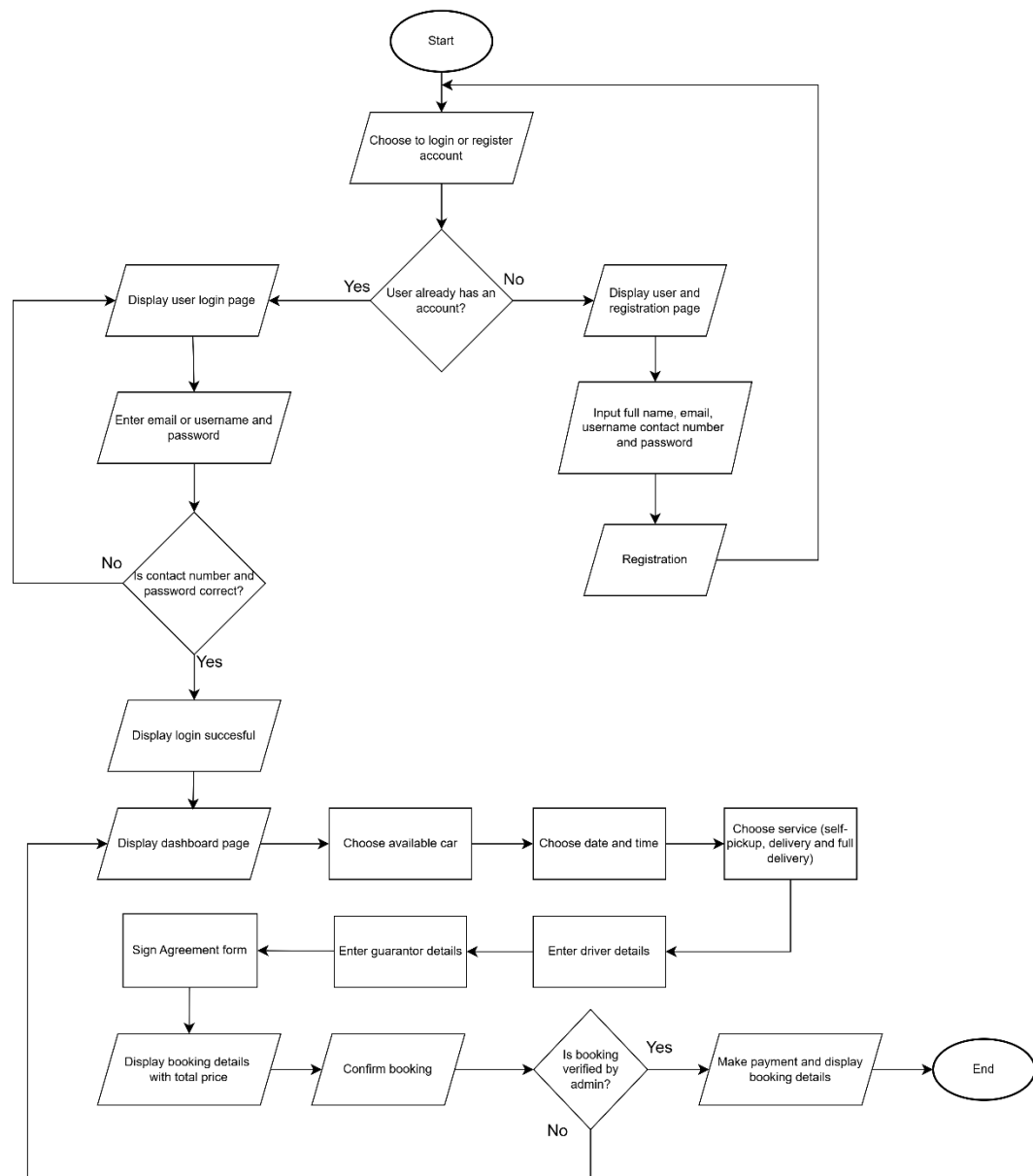


Figure 3.2 Flowchart of Proposed System

Based on Figure 3.3.1, the process begins when a customer decides to rent a car. The customer logs into the TimeLess Car Rental System and goes straight to dashboard that display the available car in real time. The customer then selects a car and fills in the required booking details, customer can choose services which is self-pickup, delivery and full delivery (pickup and return) including the driver's and guarantor's information. Once the booking request is submitted, customer have to sign the agreement form with terms and condition of TimeLess Car Rental System and it does

not go directly to payment. Instead, the booking is first reviewed by the admin (car rental owner) for verification. If the admin approves the booking, the system allows the customer to proceed to the payment stage using a secure online payment method, either by choosing a bank or entering credit card details such as card number, CVV, and expiry date. Once payment is successfully made, the system sends an automated booking confirmation email to the customer with all the booking details.

3.4 Requirement analysis of the to-be system

The requirement analysis will outline both the system requirements for each user and the functional needs.

3.4.1 Functional Requirement (Process Model)

- **User Registration and Login:** The system will provide a straightforward registration process for customers, admins, and delivery staff. Customers will be required to enter details such as their full name, email, phone number, and password during registration. Delivery staff accounts will be created and managed by the admin. Once registered, customers and delivery staff can log in using their email and password, with the system verifying their credentials against stored data. Upon successful login, users are directed to their respective dashboards. Customers can view cars and make bookings, while delivery staff can access assigned deliveries. Admins will have access to a secure login panel with full control over the system.
- **Booking Management:** Customers can view available cars in real time. Once a car is selected, customer need to choose the service which is self-pickup, delivery and full delivery include pickup and return then the customer will fill in the booking form that includes both driver and guarantor information. Next, customer need to accept and sign the terms and condition of agreement form TimeLess car rental. After submitting the booking request, it will be sent to the admin for verification. Only after approval, the customer can proceed to the payment process. Customers can also check their booking history and status updates through their dashboard. Admins can manage all bookings from their

end, including verifying, modifying, or rejecting them. This module ensures that bookings are accurate, verified, and recorded for reference.

- **Delivery Management:** Once a booking is verified and paid, the admin can assign a delivery staff member to handle the vehicle handover. Delivery staff will receive the task through their own dashboard, which shows booking details, customer info, and delivery instructions. After completing the task, delivery staff can update the delivery status. This feature ensures timely delivery coordination and gives admin visibility into the status of the handover.
- **Payment Processing:** The system integrates secure payment processing using online banking and credit card options. Customers can choose their preferred bank or enter their card number, CVV, and expiry date to complete the transaction. After successful payment, the system sends an automated booking confirmation email with all relevant details. The system also records every transaction in the database for tracking and reporting purposes. Admins can manage refunds if necessary and access a complete record of all customer payments through the dashboard.
- **Administrative Oversight:** The admin dashboard serves as the control center of the system. Admins can manage customer profiles, car inventory, bookings, driver and guarantor details, payment records, delivery staff, and system settings. They can also assign delivery staff to bookings and handle refunds. The dashboard includes built-in reporting tools to generate daily and monthly income reports, offering insights into business performance. This centralized view allows the car rental owner (admin) to maintain full control over operations and ensure the system runs smoothly.

3.4.1.1 Data Flow Diagram (DFD)

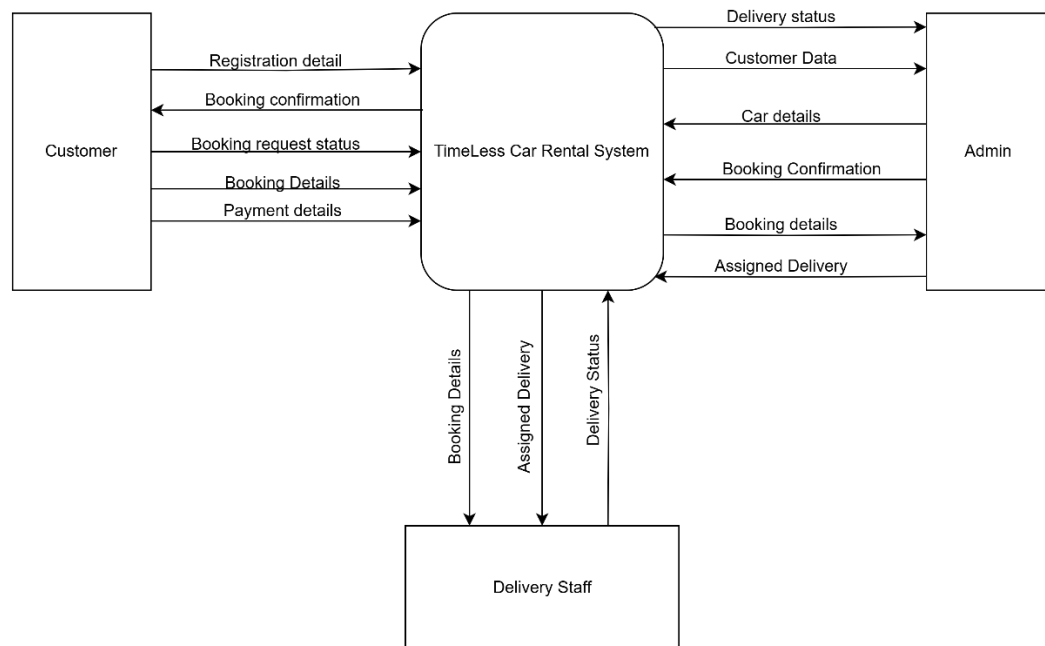


Figure 3.3 Context Diagram

As shown in Figure 3.3, the context diagram presents a high-level view of the TimeLess Car Rental System, outlining its interactions with three key external entities: Customer, Admin, and Delivery Staff. The diagram illustrates how data flows in and out of the system, establishing the system boundary and the roles each entity plays in the operation of the platform.

Customers interact with the system to perform various tasks such as account registration, booking cars, and making payments. Customers provide registration details, booking details (including driver and guarantor information), and payment information to the system. In return, the system provides booking confirmations and status updates for each request, ensuring that customers are informed about the progress of their bookings in real time.

Admins (typically the car rental business owner) are responsible for managing the core operations of the system. The admin inputs car details, views customer and booking data, and is responsible for verifying booking requests. Once a booking is approved, the admin assigns a delivery staff member through the system. The system also returns

delivery status updates and booking confirmations to the admin, allowing for continuous monitoring of bookings and deliveries.

Delivery Staff are assigned tasks by the admin through the system. They receive the booking details and assigned delivery information via their dashboard. Once the car is delivered, they update the delivery status, which is then sent back into the system and made visible to the admin and customer.

In summary, this context diagram demonstrates the main communication and data exchanges between external users and the TimeLess Car Rental System. It reflects how the system supports seamless booking, verification, payment, and delivery tracking in a centralized platform for small car rental businesses.

3.4.1.2 Data Flow Diagram (DFD) Level 0

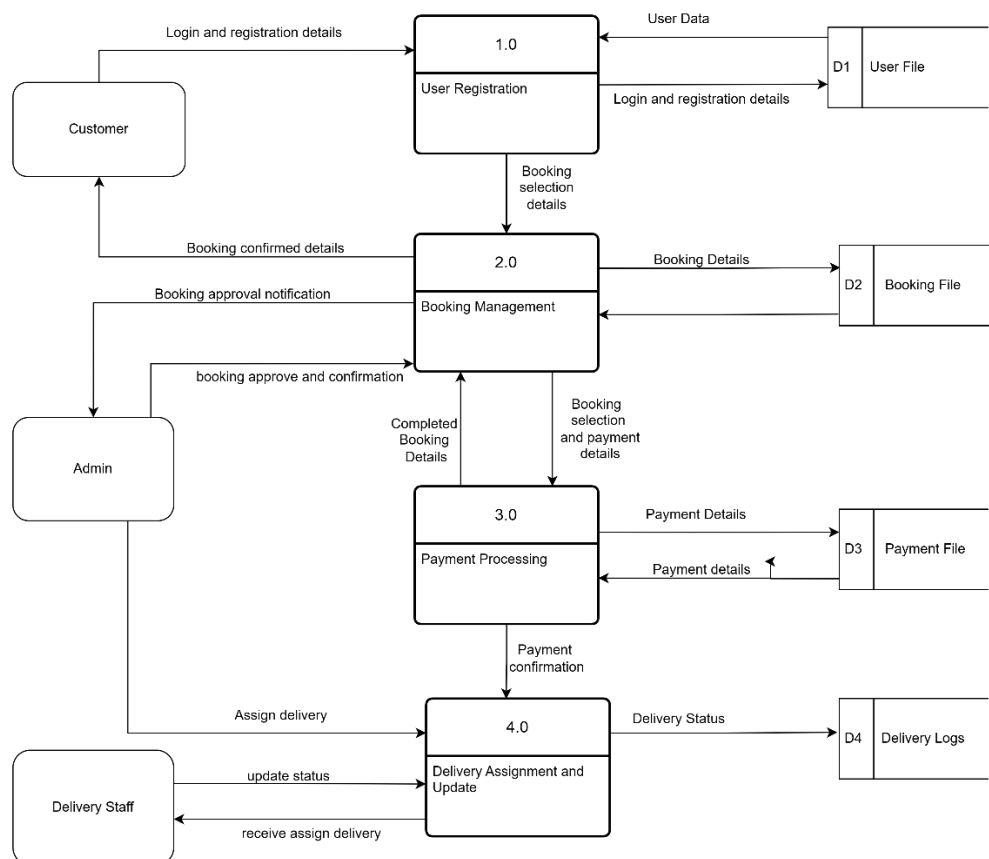


Figure 3.4 Data Flow Diagram (DFD) Level 0

According to Figure 3.4, the DFD Level 0 drawing allows us to have a high-level view of the TimeLess Car Rental System and the key functional modules of the TimeLess Car Rental System. The aim of the system is to enable smooth communication among customers, administrators and delivery personnel who have different ways of interacting with the system and flowing of data. In its essence, the system has four main modules: User Registration and Authentication, Booking Management, Payment Processing and Delivery Management.

Process 1.0: This is the User Registration and Authentication module, which customers enter in an attempt to make use of the services of the TimeLess Car Rental. The customers are expected to enter their registration data or login data and the system verifies them and stores them safely in the User File (D1). The following module allows ensuring that only the authorized users can access additional functionality of the system, whereas a secure list of all users is retained.

After the authentication, the customers get to access the Booking Management module (Process 2.0), and can check the available vehicles, create new requests, edit the existing requests or cancel the requests depending upon the policies. This module processes all the booking requests and updating of any booking request and the resultant data is stored in Booking File (D2). Such a mechanism guarantees the possibility of the up-to-date data in terms of booking and booking status to both customers and administrators.

Payment Processing module (Process 3.0) is controlled mostly by the administrator, who has to verify financial operations made and record the payments when it comes to booking. These are receiving payment of customers and canceling refund where a customer is entitled, and making certain that payment details are regularly updated in the Payment File (D3). Combining this module with the rest of the system allows the secure and transparent transactions of all the financial affairs covering the rental process.

The Delivery Management system (Process 4.0) performs an assignment and follow-up of vehicle delivery and returns. This module is used to assign tasks to the delivery staff and inform the system about the status of every operation of the delivery process.

All the pertinent information and data such as time stamps and completion information will be captured in the Delivery logs (D4) and will provide an in-depth tracking and accountability of delivery lifecycle.

In conclusion, the DFD Level 0 diagram on TimeLess Car Rental System helps to show the key processes in the system, data flow between the system and external entities and the way key data files are organized. All the modules serve a specific purpose to self-stable performance, unquestionably safe, and optimal operations of the system to support user experience and the administrative management of the system.

3.4.2 Non-Functional Requirement

Non-functional requirements are used to specify the qualities and properties of a system that guarantee overall efficiency, usability and also reliability of such a system. The requirements are essential to provide a system that may not only execute desired functions, but also satisfy the user expectations regarding its performance, security, and provision of the user experience. Non-functional requirements of TimeLess Car Rental System will be divided into two broad points that are quality requirements and performance requirements.

3.4.2.1 Quality Requirements

- **Accuracy:** It should be accurate to eliminate the possibility of making any mistake in the running of all the car rental operations. All user information such as the customer registration records, car availability, to the booking, the records of the drivers and those of the guarantors, and the records of the payments should be captured and processed accurately by the system. As an example, on the moment a customer reserves a car, the system should be able to properly set the car that is selected, the time of vehicle rental, a driver and a guarantor. Correct amount of payment must be made in regards to the rental rate with any form of services applied. To encourage this, the system will introduce validations done with great strength where completeness of data as well as avoiding repetition of data and also verification that there is consistency in

aggregation of data among the modules comes in hand; hence, minimizes disputes or errors which can occur because of incorrect treatment of data.

- **Security:** The TimeLess Car Rental System pays special attention to security because it stores sensitive user info and online financial operations. The system will have secure log-on details to all users and will have a role-based access so as to avoid intrusion actions. All passwords will be encrypted using high levels of encryption and all the data between the users and the system will be encrypted using SSL/TLS encryption. The protection measures to be implemented to counter the common threats in the system will encompass input sanitization to guard against SQL injection and cross-side scripting (XSS) filters. Before the payment can be processed customers will undergo an additional layer of security through booking steps verified by the administrator. This will keep users and their data safe given that regular security checks and firewalls will keep at bay the latest vulnerabilities.
- **Reliability:** The reliability plays an important role to maintain the availability and excellent functioning of the system. TimeLess Car Rental System must be functional without time out in order to allow the users to make booking, payment and delivery services available at any time. The system architecture will be developed to support the environment of a large number of concurrent users and booking requests without going down or equipment overload. Automated backups will be undertaken to avoid wastage of data and other related problems that threaten the survival of the firm. There will be recovery mechanism to re-establish the system in the event of data corruption or server failure. Such precautions help fast track the system to retrieve any problems and offer constant service to customers as well as the admins.
- **Usability:** The system should provide an intuitive interface that should be simple to use by any end user whether customers, admins as well as delivery staff. A clear labeling, logical navigation and consistency layout of interface will be used. As an example, customers need to make few clicks to find cars, enter the data about the drivers and guarantors, and make reservations easily. Delivery workers will be able to view the simplified dashboard and view the

assigned tasks, and admins will be presented with the management modules of cars, users, bookings, payments, and reports. It will be a responsive design and there will be no loss of functionality on desktops, tablets, and smartphones. When testing and deploying the software, usability feedback will be gathered so that future versions can be improved and that the system would be easy to use by users with different backgrounds.

3.4.2.2 Performance Requirements

- **Response Time:** TimeLess Car Rental System that will create a responsive and enjoyment user experience is affected by the response time. The system should have the capacity to give quick results to user operations like search car, booking cars, pay and delivery updates. An example would be when a customer makes a search inquiring about available vehicles or sets a booking through driver and guarantor details, a system should be able to give results and update the database in few seconds. In the same way, during each booking confirmation or when the admin makes a monthly report of the income, the system must accomplish the tasks without any apparent lag. Such rapid response will be achieved by properly optimized code, optimally indexed database queries and proper management of user sessions. This will guarantee that every user such as the customers, the admins, and the delivery staff can communicate with the system, without delay or frustration.
- **Scalability:** Scalability is vital in that it allows the maintenance of future increases in several users, cars, and other features of the TimeLess Car Rental System. As a platform attracts more customers, the number of bookings that go through the system increases, and more branches or delivery people are introduced, the system must be scalable without decreasing in performance. Due to the dynamic nature of the task, the system will be modeled in a modular way in order to ensure it is easier to add new elements to the system (payment gateways variant or reporting variety etc.) without disruption to current functioning. In the long-term, the platform may be moved to the cloud infrastructure to enable the dynamic allocation of resources, so that the system will be stable even in the hours of extreme traffic. When designing a scalable

system, it is designed to accommodate the increase in business to ensure that the system still delivers to the various business needs.

- **Capacity:** The software has to be able to support an excessive amount of information and parallel users particularly at rush hours. These are the storage of customer profiles, booking details, driver and guarantor details, payment details, and delivery details. The database shall have high performance read/write operations so the database will work well even in heavy loading. The system must also be capable of handling concurrent customers who could book cars without compromising its speed and reliability. Load testing will be regularly done to determine the shortages in the current infrastructure so that steps can be taken in advance. Moreover, sufficient storage and server power will be in place and they can smoothly be upscaled in case the volume of data increases. Adequate capacity enables the system to have stable performance and availability when business growth is achieved.

3.4.3 Other Requirement

The other requirements in a project specification involve the different criteria besides the software and hardware requirements. These usually include performance measures, including system the response time and efficiency and the security measures to protect data. safe access. It is essential to adhere to the applicable rules and norms, in addition to usability requirements of ease of use and accessibility. Reliability and availability expectations deal with system up-time and fault tolerance. maintainability criteria make the system easily updatable and repairable. The ability to work with other systems and software as well as the necessity of sufficient training and documentation also comes under this category.

3.4.3.1 Software Requirements

Table 3.1 Software Requirements

Software	Benefit	Reason

Microsoft Visual Studio Code	Visual Studio Code is a light-weight but potent source code editor that runs on your desktop and it is available Windows, macOS and Linux. It is accompanied by in-built support of JavaScript and boasts of an ecosystem other language extension	Visual Studio Code is a powerful and comfortable coding, debugging and version control environment.
XAMPP	XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, and interpreters for scripts written in the PHP and Perl programming languages.	XAMPP provides an easy-to-use control panel to manage your Apache Server, MySQL databases, and PHP scripts.

3.4.3.2 Hardware Requirements

Table 3.2 Hardware Requirements

Hardware	Specification	Reason
Laptop HP	Windows 11 Home Single Language	Windows 11 is the latest operating system from Microsoft, offering improved performance, security, and a user-friendly interface
	12th Gen Intel(R) Core (TM) i5-1235U 1.30 GHz	The Intel Core i5 processor provides sufficient processing power for developing and testing your system. It can handle multiple tasks and processes without lagging, ensuring a smooth development experience.
	8.00 GB (7.68 GB usable)	4GB of RAM is enough for running the necessary software for development like Visual Studio Code, XAMPP, and web

		browsers for testing. It allows for efficient multitasking and quick access to data and applications.
	64-bit operating system, x64-based processor	A 64-bit operating system can handle large amounts of memory more efficiently than a 32-bit operating system. This is beneficial for running applications that require a lot of memory, such as database management systems and web servers.

3.5 Conclusion

In conclusion, the chapter has carried out an analysis of the prevailing car rental procedure, has ascertained its flaws, followed by a proposed solution a new web-based system, that will overcome these flaws. The given examples of the functional and non-functional requirements of the proposed TimeLess Car Rental System entail high details, and the main targets of using them concern automating the booking process, enhancing the control of the admin, and the overall user experience of the customers and delivery staff. Software specifications have also been described giving requirements of the tools and technologies that will be required to develop and deploy the software. The benefits of this analysis lay in the good basis of the design and implementation of the system which would enhance efficiency, accuracy, and professionalism of the small-scale car rental operations.

CHAPTER 4: DESIGN

4.1 Introduction

This chapter describes the car rental system architecture and the design of the database used management system. It gives a detailed overview of architecture of the system, such as layers, structures and design patterns, after conducting an in-depth discussion of the database design process, including the conceptual, logical and physical design phases. The chapter ends by the design and description of the graphical user interface (GUI), so as to be consistent with the non-functional and functional requirements.

4.2 System Architecture

The TimeLess Car Rental System is developed using architecture that would deliver scalable, secure, and efficient platform to perform the car rental business. It implements a three-tier architecture which is composed of the presentation layer, the operational layer and data layer.

- **Presentation Layer:** This layer represents the user interface and interactions for all types of users, customer, admin, and delivery staff. It is accessed via a web browser and is developed using HTML, CSS, and JavaScript, providing a responsive and user-friendly experience across different devices. Customers can register, book cars, and make payments, while admins manage bookings, vehicles, and staff, and delivery staff view assigned tasks and update delivery statuses.
- **Operational Layer:** This middle layer handles the core functions of the system. It includes user authentication, booking management, payment processing, delivery tracking, and report generation. The business logic is implemented using PHP, with support from PHPMailer for sending email confirmations. This layer also ensures that admins can approve bookings, manage records, and monitor system activity through a secure dashboard.
- **Data Layer:** This layer is responsible for data storage, retrieval, and management. The system uses MySQL with phpMyAdmin to handle data related to customers, vehicles, bookings, payments, drivers, guarantors, and delivery records. The database is designed with normalized tables to ensure data integrity and reduce redundancy.

4.3 Database Design

The database design process includes three stages namely conceptual design, logical design and physical design. The goal of each of the phases is to make sure that the database structure fits the needs and goal of the operation of the system.

4.3.1 Conceptual Design

4.3.1.1 Entity Relationship Diagram (ERD)

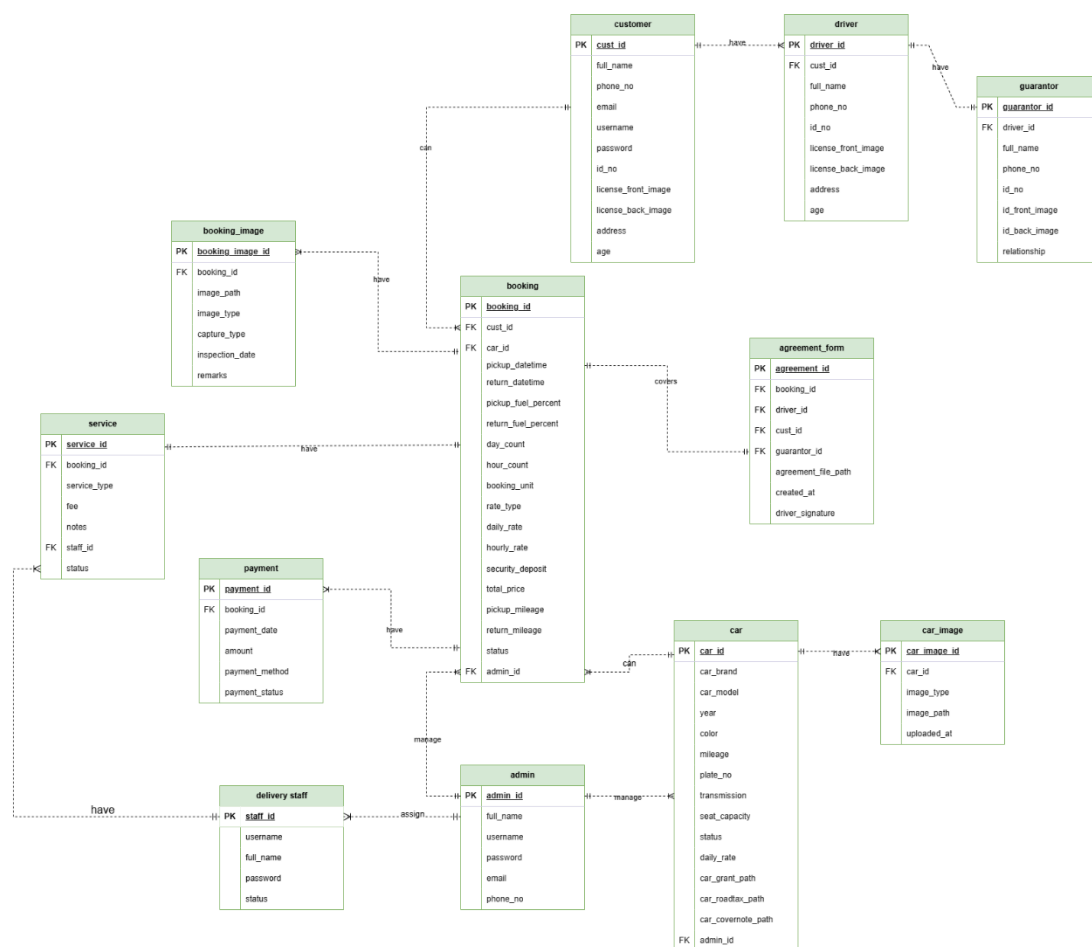


Figure 4.1 ERD

4.3.1.2 Business Rules

- I. Each customer can make many bookings.
- II. Each driver is linked to customer, but not all customers are drivers.

- III. Each guarantor is assigned to one driver only.
- IV. Each car can be booked multiple times, but only by one customer per booking.
- V. Each car can have multiple display images, uploaded by an admin.
- VI. A booking can have multiple booking images to record the car's condition before and after the rental.
- VII. A booking can only have one agreement form.
- VIII. Each booking image belongs to one booking and must include image type, capture type (pickup or return), and upload time.
- IX. Each delivery staff may be assigned to multiple services
- X. An admin manages bookings and maintains car details and pricing.
- XI. Each car is managed by one admin only.
- XII. A booking can request one services, and service is handled by one delivery staff.

4.3.2 Logical Design

The logical design of database is the second step that is involved in the process of database design. At this stage, connectivity is set between nearby local logical data and normalized and tested with user transactions. Also, the limitations of every local logical information are checked.

4.3.2.1 Data Dictionary for Entity Relationship Diagram

Table 4.1 Admin Table

ADMIN

Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
admin_id	admin id	int (11)	No	-	Yes	PK	-
full_name	admin full name	varchar (100)	No	-	No	-	-
username	admin username	varchar (50)	No	-	No	-	-
password	admin password	varchar (255)	No	-	No	-	-
email	admin email	varchar (100)	No	-	No	-	-
phone_no	admin phone number	varchar (20)	Yes	Null	No	-	-

Table 4.2 Delivery Staff Table

DELIVERY_STAFF							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
staff_id	delivery staff id	int (11)	No	-	Yes	PK	-
full_name	delivery staff full name	varchar (100)	No	-	No	-	-
username	delivery staff username	varchar (50)	No	-	No	-	-

password	delivery staff password	varchar (255)	No	-	No	-	-
status	status of the delivery staff (active or inactive)	enum ('active', 'inactive')	No	-	No	-	-

Table 4.3 Customer Table

CUSTOMER							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
cust_id	customer id	int (11)	No	-	Yes	PK	-
full_name	customer full name	varchar (100)	No	-	No	-	-
username	customer username	varchar (50)	No	-	No	-	-
password	customer password	varchar (255)	No	-	No	-	-
email	customer email	varchar (100)	No	-	No	-	-

phone_no	customer phone number	varchar (20)	Yes	Null	No	-	-
address	customer address	text	Yes	Null	No	-	-
age	customer age	int (11)	Yes	Null	No	-	-
id_no	customer identification number	varchar (50)	No	-	No	-	-
license_front_image	customer license front image	longblob	No	-	No	-	-
license_back_image	customer license back image	longblob	No	-	No	-	-

Table 4.4 Driver Table

DRIVER							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
driver_id	driver id	int (11)	No	-	Yes	PK	-
full_name	driver full name	varchar (100)	No	-	No	-	-

id_no	driver identification number	varchar (50)	No	-	No	-	-
license_front_image	driver license front image	longblob	No	-	No	-	-
license_back_image	driver license back image	longblob	No	-	No	-	-
phone_no	driver phone number	varchar (20)	No	-	No	-	-
address	driver address	text	Yes	Null	No	-	-
age	driver age	int (11)	Yes	Null	No	-	-
cust_id	customer id	int (11)	No	-	Yes	FK	customer

Table 4.5 Booking Table

BOOKING							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
booking_id	booking id	int (11)	No	-	Yes	PK	-
pickup_datetime	booking pickup date and time	datetime	Yes	Null	No	-	-

return_datetime	booking return date and time	datetime	Yes	Null	No	-	-
day_count	booking day count	int (11)	Yes	0	No	-	-
hour_count	booking hour count	int (11)	Yes	0	No	-	-
booking_unit	unit of the booking like hour or day	enum ('hour', 'day')	Yes	day	No	-	-
rate_type	type of rate applied to the booking	enum ('hourly', 'daily')	Yes	daily	No	-	-
daily_rate	The daily rental price applied to the booking	decimal (10,2)	Yes	Null	No	-	-
hourly_rate	The hourly rental price applied to the booking	decimal (10,2)	Yes	Null	No	-	-
total_price	Total price of the booking	decimal (10,2)	Yes	Null	No	-	-
security_deposit	security deposit per booking	decimal (10,2)	No	100.00	No	-	-
pickup_mileage	the mileage before pickup	int (11)	Yes	Null	No	-	-
return_mileage	the mileage after return	int (11)	Yes	Null	No	-	-
status	booking status	enum('pending', 'confirmed',	Yes	Null	No	-	-

		'completed', 'cancelled', 'waiting — verification', 'rejected', 'approved')					
created_at	booking created	time stamp	No	-	No	-	-
pickup_fuel_percent	fuel percent before pickup	int (11)	Yes	Null	No	-	-
return_fuel_percent	fuel percent after return	int (11)	Yes	Null	No	-	-
rejection_reason	rejection reason of booking	varchar (255)	Yes	Null	No	-	-
cust_id	customer id	int (11)	No	-	Yes	FK	customer
driver_id	driver id	int (11)	No	-	Yes	FK	driver
car_id	car id	int (11)	No	-	Yes	FK	car

Table 4.6 Booking Image Table

BOOKING IMAGE							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
booking_image_id	booking image id	int (11)	No	-	Yes	PK	-
image_path	booking image	longblob	Yes	Null	No	-	-

	file path						
image_ type	booking image type like car side.	varchar (50)	Yes	Null	No	-	-
capture_ type	type of booking image pickup or return	varchar (50)	Yes	Null	No	-	-
inspection _date	inspection date and time	datetime	Yes	Null	No	-	-
remarks	remarks of the booking	text	Yes	Null	No	-	-
booking_ id	booking id	int (11)	No	-	Yes	FK	booking

Table 4.7 Car Table

CAR							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
car_id	car id	int (11)	No	-	Yes	PK	-
car_brand	brand of the car	varchar (50)	Yes	Null	No	-	-

car_model	model of the car	varchar (50)	Yes	Null	No	-	-
year	year of the car	int (11)	Yes	Null	No	-	-
color	color of the car	varchar (30)	Yes	Null	No	-	-
mileage	mileage of the car	int (11)	Yes	Null	No	-	-
plate_no	car plate number	varchar (20)	Yes	Null	No	-	-
transmission	car transmission	varchar (20)	Yes	Null	No	-	-
seat_capacity	seat capacity of the car	int (11)	Yes	Null	No	-	-
status	status of the car	enum ('available', 'not available')	Yes	available	No	-	-
daily_rate	rate of the car	decimal (10,2)	Yes	Null	No	-	-
car_grant_path	car grant	longblob	Yes	Null	No	-	-
car_roadtax_path	car roadtax	longblob	Yes	Null	No	-	-
car_cover_note_path	car covernote	longblob	Yes	Null	No	-	-
hourly_rate	hourly rate of the car	decimal (10,2)	Yes	8.00	No	-	-
admin_id	admin id	int (11)	No	-	Yes	FK	admin

Table 4.8 Car Image Table

CAR IMAGE							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
car_image_id	car image id	int (11)	No	-	Yes	PK	-
image_path	car image file path	longblob	Yes	Null	No	-	-
image_type	car image type like 'main'	varchar (50)	Yes	Null	No	-	-
uploaded_at	car image upload time and date	datetime	Yes	Null	No	-	-
car_id	car id	int (11)	No	-	Yes	FK	car

Table 4.9 Agreement Form Table

AGREEMENT_FORM							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
agreement_id	agreement id	int (11)	No	-	Yes	PK	-

agreement_file_path	agreement file	longblob	Yes	Null	No	-	-
driver_signature	driver signature	longblob	Yes	Null	No	-	-
booking_id	booking id	int (11)	No	-	Yes	FK	booking
driver_id	driver id	int (11)	No	-	Yes	FK	driver
cust_id	customer id	int (11)	No	-	Yes	FK	customer
guarantor_id	guarantor id	int (11)	No	-	Yes	FK	guarantor

Table 4.10 Guarantor Table

GUARANTOR							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
guarantor_id	guarantor id	int (11)	No	-	Yes	PK	-
full_name	guarantor full name	varchar (100)	No	-	No	-	-
id_no	guarantor identification number	varchar (50)	No	-	No	-	-
id_front_image	guarantor license front image	longblob	No	-	No	-	-
id_back_image	guarantor license back image	longblob	No	-	No	-	-
phone_no	guarantor phone number	varchar (20)	No	-	No	-	-

relation-ship	guarantor	varchar (50)	Yes	-	No	-	-
driver_id	driver id	int (11)	No	-	Yes	FK	driver

Table 4.11 Payment Table

PAYMENT							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
payment_id	payment id	int (11)	No	-	Yes	PK	-
payment_date	payment date	date	Yes	Null	No	-	-
amount	total amount of the booking	decimal (10,2)	Yes	Null	No	-	-
payment_method	payment method	varchar (50)	Yes	Null	No	-	-
payment_status	payment status	enum ('pending', 'paid')	Yes	Null	No	-	-
booking_id	booking id	int (11)	No	-	Yes	FK	booking

Table 4.12 Service Table

SERVICE							
Attribute Name	Description	Data type and size	Null	Default Value	Unique	PK or FK	Reference table
service_id	service id	int (11)	No	-	Yes	PK	-

service_ type	service type, delivery, self-pickup or full delivery	varchar (100)	Yes	Null	No	-	-
fee	fee of each service	decimal (10,2)	Yes	Null	No	-	-
notes	location to delivered the car	text	Yes	Null	No	-	-
status	status of the delivery	enum (‘pendin g’,out for delivery’ ,‘delivere d’)	Yes	pending	No	-	-
staff_id	staff id	int (11)	No	-	Yes	FK	delivery _staff

4.3.2.2 Query Design

There are many query designs that can be used to create different types of outcomes. Each query has its own requirements, reasons and purpose.

Table 4.13 SQL Table

Type Of SQL	SQL statement	Explanation
Simple sql	SELECT * FROM customer;	Retrieve all customer information
Join Table	SELECT b.booking_id, b.pickup_datetime, b.return_datetime, b.total_price, c.full_name AS customer_name, car.car_brand, car.car_model FROM booking b JOIN customer c ON b.cust_id = c.cust_id JOIN car ON b.car_id = car.car_id;	This query retrieves booking information along with related customer and car details.
Aggregate and order	SELECT b.*, c.car_brand, c.car_model, c.plate_no, ci.image_path AS car_image FROM booking b JOIN car c ON b.car_id = c.car_id LEFT JOIN (SELECT car_id, image_path FROM car_image WHERE car_image_id IN (SELECT MIN(car_image_id) FROM car_image GROUP BY car_id)) ci ON c.car_id = ci.car_id WHERE b.cust_id = ? ORDER BY b.pickup_datetime DESC;	This query retrieves a list of bookings for a specific customer, along with detailed car information and one representative image for each car (the earliest image uploaded).

4.4 Graphical User Interface (GUI) Design

4.4.1 Customer Interface

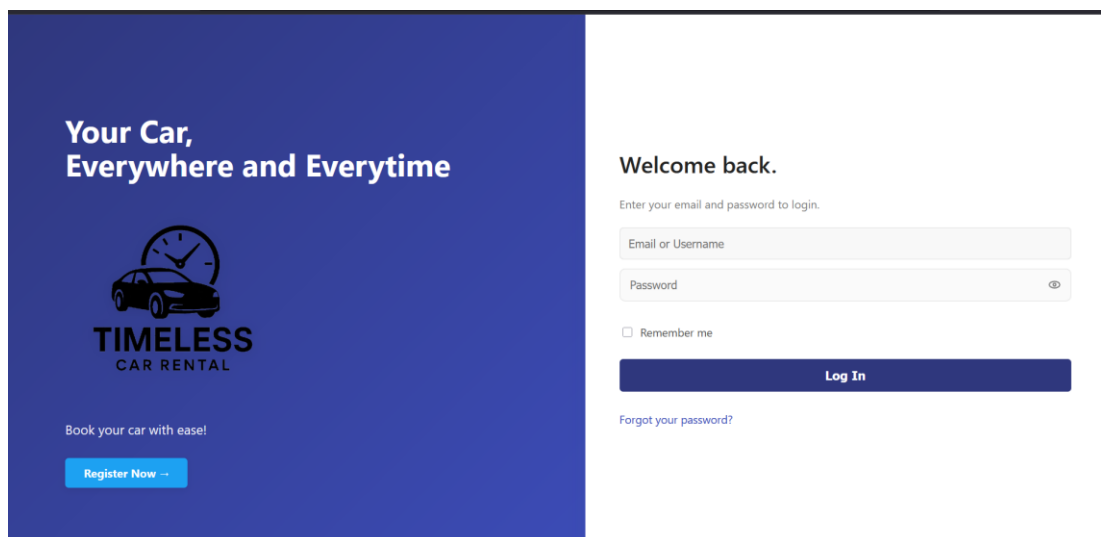


Figure 4.2 Customer Login Page

As shown in Figure 4.1, this page provides a login interface specifically for customers of the Timeless Car Rental system. It features a modern, user-friendly design that allows users to sign in using their email or username and password to access their accounts. The left section displays the company's branding, a promotional tagline, and a call-to-action button for new user registration, while the right section contains the login form with options such as "Remember me" and password recovery.

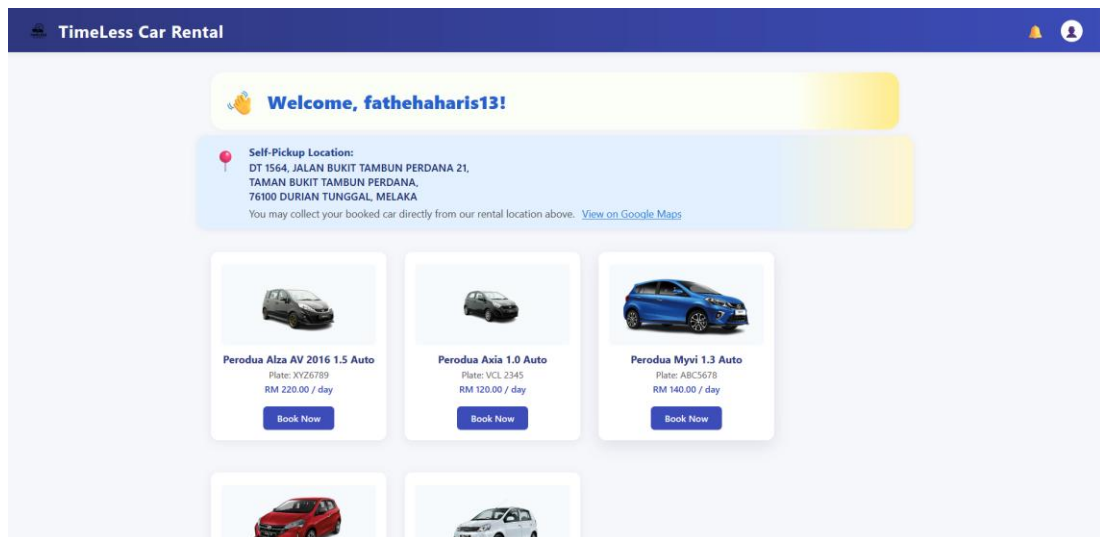


Figure 4.3 Customer Dashboard Page

As shown in Figure 4.2, this page serves as the main dashboard for a logged-in customer of the TimeLess Car Rental system. The top section features a personalized welcome message addressing the user by their username. Below the welcome message, the page displays the self-pickup location for car rentals, including the full address and a link to view the location on Google Maps.

The main content area showcases a selection of available cars for rent, presented as individual cards. Each card displays the car's image, model, plate number, daily rental rate, and includes a "Book Now" button for users to proceed with their reservation. The layout is clean and organized, making it easy for customers to browse and select a car for rental.

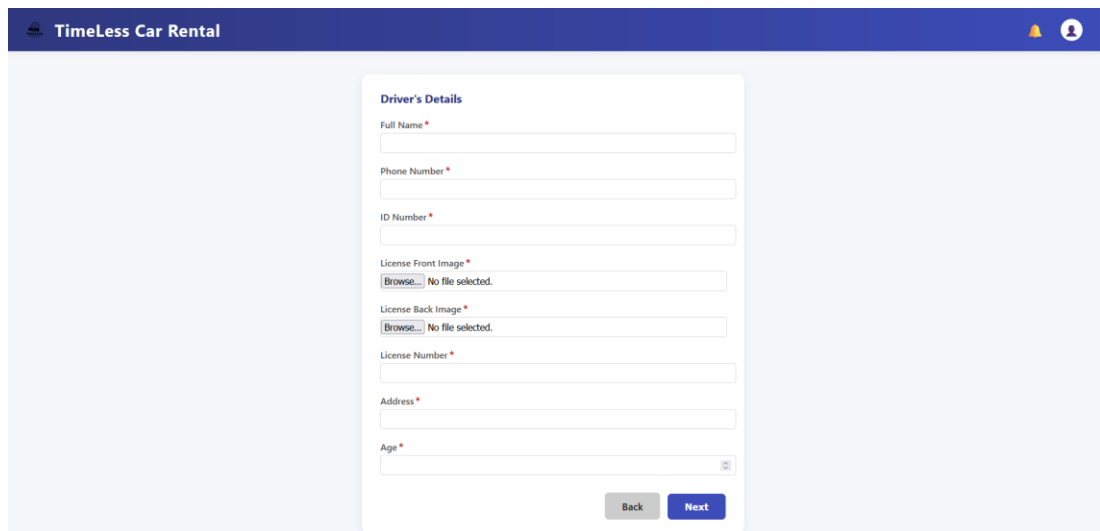
The screenshot displays the 'TimeLess Car Rental' web application. At the top, a dark blue header contains the company name and a user profile icon. The main content area features a white card with the following details:

- Car Model:** Perodua Myvi 1.3 Auto
- Image:** A blue Perodua Myvi 1.3 Auto.
- Specifications:**
 - Plate No: ABC5678
 - Year: 2015
 - Color: Blue
 - Transmission: Auto
 - Seat Capacity: 5
 - Mileage: 97000 km
 - Daily Rate: RM 140.00
 - Hourly Rate: RM 8.00
- Booking Fields:**
 - Pickup Date:** A date input field followed by a time dropdown menu set to '00:00'.
 - Return Date:** A date input field followed by a time dropdown menu set to '00:00'.
 - Service:** A dropdown menu currently showing 'Pick up myself (FREE)'.
- Navigation:** 'Back' and 'Next' buttons at the bottom of the card.

Figure 4.4 Book Car Page

As shown in Figure 4.3, this page presents the car booking details interface for the TimeLess Car Rental system. The section at the center of the page displays comprehensive information about the selected car, including the model, an image of the vehicle, plate number, year, color, transmission type, seat capacity, mileage, daily and hourly rates.

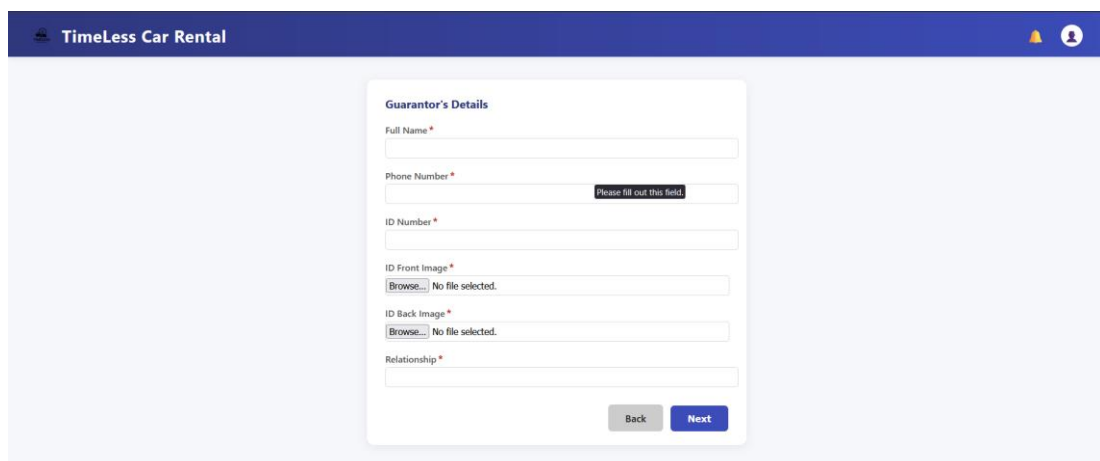
Below the car details, the interface provides input fields for the user to specify the pickup and return dates and times, as well as a dropdown menu to select the preferred service option (self-pickup, delivery and full delivery). Navigation buttons labeled "Back" and "Next" are provided at the bottom, allowing users to move through the booking process. The layout is clear and organized, enabling users to easily review car details and proceed with their reservation.



The screenshot shows the 'Driver's Details' form within the 'TimeLess Car Rental' application. The form is centered on a light blue background. It contains the following fields: 'Full Name *', 'Phone Number *', 'ID Number *', 'License Front Image *' (with a 'Browse...' button and 'No file selected.' text), 'License Back Image *' (with a 'Browse...' button and 'No file selected.' text), 'License Number *', 'Address *', and 'Age *' (with a dropdown arrow). At the bottom right of the form are 'Back' and 'Next' buttons. The application header is dark blue with the logo and name 'TimeLess Car Rental' on the left, and a bell icon and user profile icon on the right.

Figure 4.5 Driver Details Page

As shown in Figure 4.4, this page provides the interface for entering driver's details in the TimeLess Car Rental system. The form collects essential information required for driver verification, including full name, phone number, ID number, license number, address, and age. It also includes file upload fields for the front and back images of the driver's license, ensuring that proper documentation is attached. All fields marked with an asterisk (*) are mandatory, ensuring that users provide complete and accurate information before proceeding.



The screenshot shows the 'Guarantor's Details' form within the 'TimeLess Car Rental' application. The form is centered on a light blue background. It contains the following fields: 'Full Name *', 'Phone Number *' (with a 'Please fill out this field.' tooltip), 'ID Number *', 'ID Front Image *' (with a 'Browse...' button and 'No file selected.' text), 'ID Back Image *' (with a 'Browse...' button and 'No file selected.' text), and 'Relationship *'. At the bottom right of the form are 'Back' and 'Next' buttons. The application header is dark blue with the logo and name 'TimeLess Car Rental' on the left, and a bell icon and user profile icon on the right.

Figure 4.6 Guarantor's Details Page

As shown in Figure 4.5, this page presents the guarantor's details form in the TimeLess Car Rental system. The form is designed to collect important information about the

guarantor, including their full name, phone number, ID number, and the nature of their relationship to the renter. Additionally, there are fields for uploading images of the front and back of the guarantor's identification card, ensuring that all necessary documentation is provided.

TimeLess Car Rental

Review & Confirm Your Booking

Car & Booking Details

Car	Perodua Myvi 1.3 Auto
Rental Type	Daily
Daily Rate	RM 140.00
Daily Count	1 daily(s)
Pickup	2025-07-01 08:00:00
Return	2025-07-02 08:00:00
Delivery Type	Delivery
Delivery Fee	RM 10.00
Delivery Location	SC 1
Subtotal	RM 140.00
Security Deposit	RM 100.00
Total Amount	RM 250.00

Driver (Customer)

Name	Nur Fathehah Binti Mohd Aris
Phone	0199590828
ID No	020313081138
License No	020313081138
Address	10, haha

Figure 4.7 Review Booking Page

Phone	0199590828
ID No	020313081138
License No	020313081138
Address	10, haha
Age	23
License Front Image	
License Back Image	

Guarantor

Name	Mohd Aris Bin Zainuddin
Phone	0195039128
ID No	660913065081
Relationship	father
ID Front Image	
ID Back Image	

Back **Next**

Figure 4.8 Review Booking Page

As shown in Figure 4.6 and 4.7, this page displays the “Review & Confirm Your Booking” summary in the TimeLess Car Rental system. The interface presents a comprehensive overview of all booking details before final submission. This structured summary ensures users can verify all provided information and uploaded

documents, reducing the risk of errors before completing their booking. The clear layout and categorization enhance the user experience by making it easy to review each component of the booking process.

TimeLess Car Rental

Rental Agreement & Signature

AGREEMENT OF VEHICLE USAGE BETWEEN BORROWER AND TIMELESS CAR RENTAL

TimeLess Car Rental is a brand operated by TimeLess Car Rental. Attached herewith are the terms and conditions that shall be between the BORROWER of the vehicle and TimeLess Car Rental. When the agreement is signed, the BORROWER has acknowledged that he/she has read, understood and agreed to the terms and conditions.

IT IS HEREBY AGREED AS FOLLOWS:

☒ I have read and agree to all terms and conditions above.

Signature (draw below):

[Handwritten Signature]

Figure 4.9 Rental Agreement and Signature Page

This figure 4.8, captures the **digital rental agreement and customer signature**, ensuring that users formally acknowledge the terms and conditions of vehicle usage before finalizing a booking.

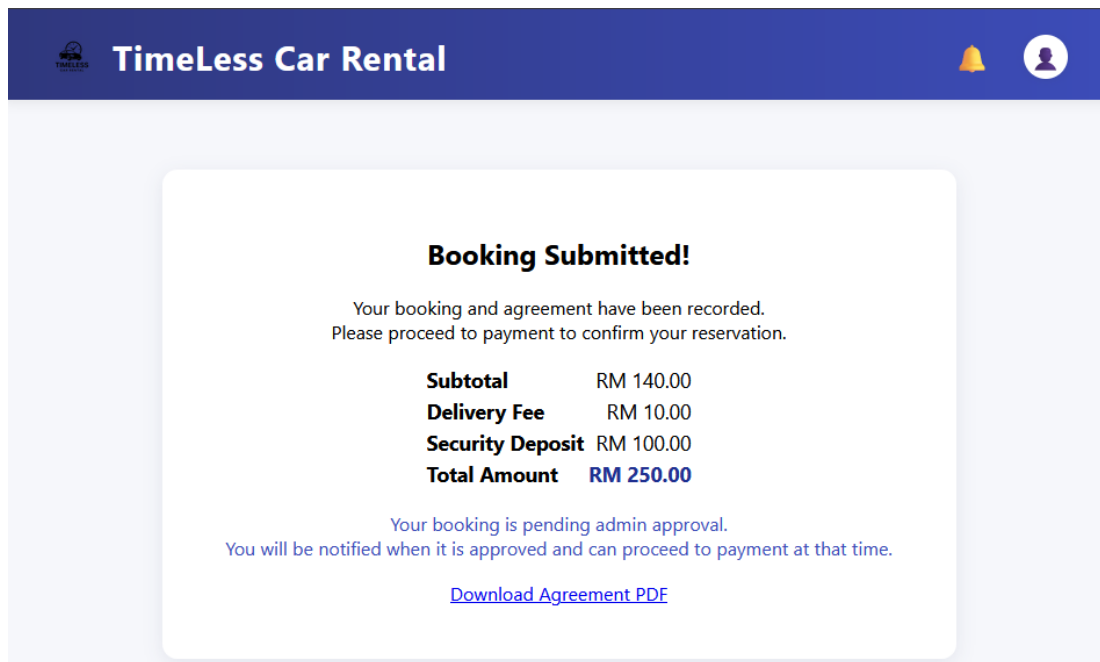


Figure 4.10 Booking Submission Page

Figure 4.9 confirms that the customer's car booking and rental agreement have been successfully submitted. It displays a clear breakdown of charges including the subtotal, delivery fee, and security deposit, with a total amount to be paid. The system informs the user that the booking is pending admin approval, and payment can only proceed once it is approved. A link is provided for the user to download the signed agreement in PDF format for reference.

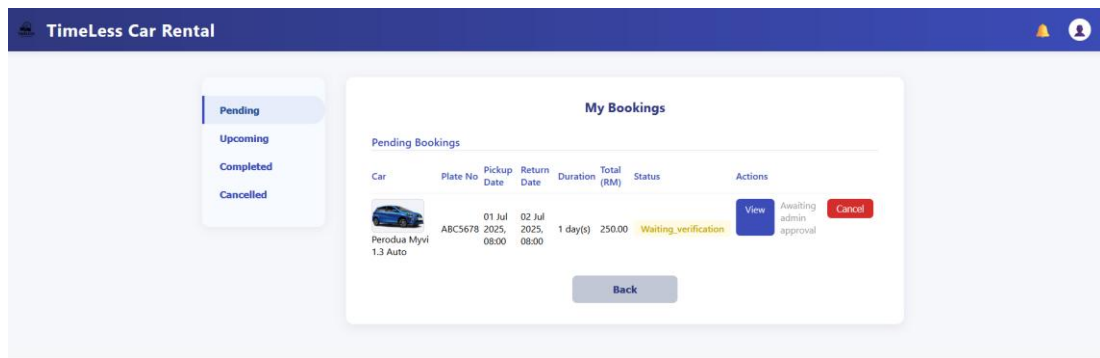


Figure 4.11 Customer Booking Page

Figure 4.10 displays the "Pending Bookings" section within the customer's My Bookings page on the TimeLess Car Rental system. It shows details of a booking that has been submitted but is still awaiting admin approval. Information such as the car model, plate number, pickup and return dates, rental duration, and total amount are clearly listed. The booking status is marked as "Waiting_verification", and the system notes that it is "Awaiting admin approval". Customers have the option to view booking details or cancel the booking if needed. This feature helps users easily track the progress and status of their reservation.

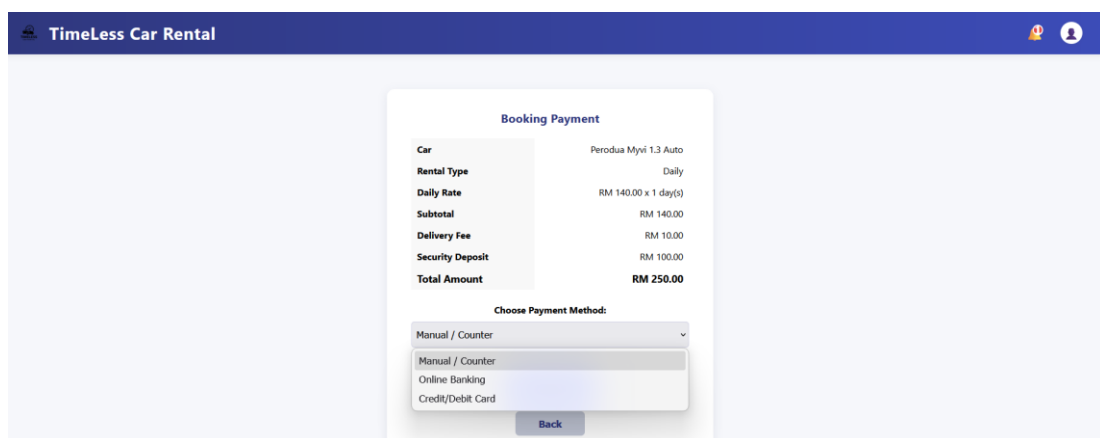
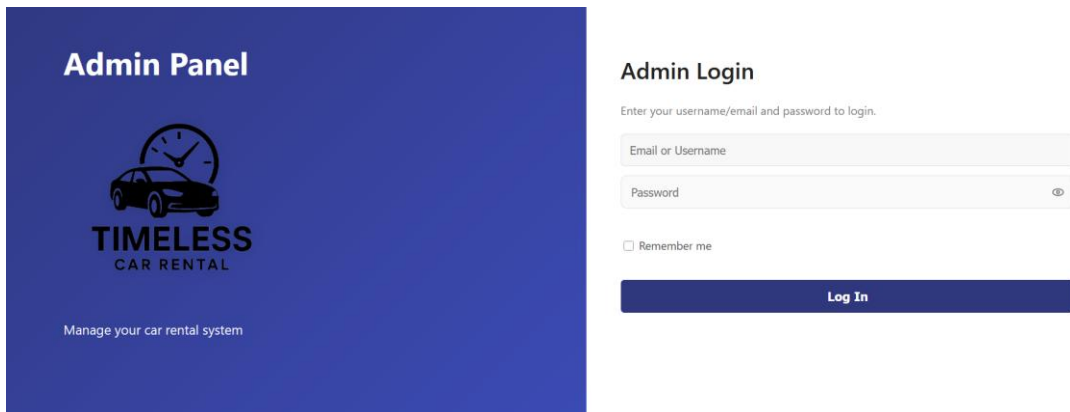


Figure 4.12 Payment Page

This figure 4.11 shows the payment section of the Timeless Car Rental system, where customers finalize their booking by selecting a payment method.

4.4.2 Admin Interface



The image shows the Admin Login Page for the Timeless Car Rental system. It is divided into two main sections. The left section, titled 'Admin Panel', features the company logo (a car inside a clock) and the text 'TIMELESS CAR RENTAL' and 'Manage your car rental system'. The right section, titled 'Admin Login', contains a login form with fields for 'Email or Username' and 'Password', a 'Remember me' checkbox, and a 'Log In' button. A small instruction 'Enter your username/email and password to login.' is above the fields.

Figure 4.13 Admin Login Page

As shown in Figure 4.1, this page provides a login interface specifically for admin of the Timeless Car Rental system. It features a modern, user-friendly design that allows users to sign in using their email or username and password to access their accounts. The left section displays the company's branding, a promotional tagline, while the right section contains the login form with options such as "Remember me".

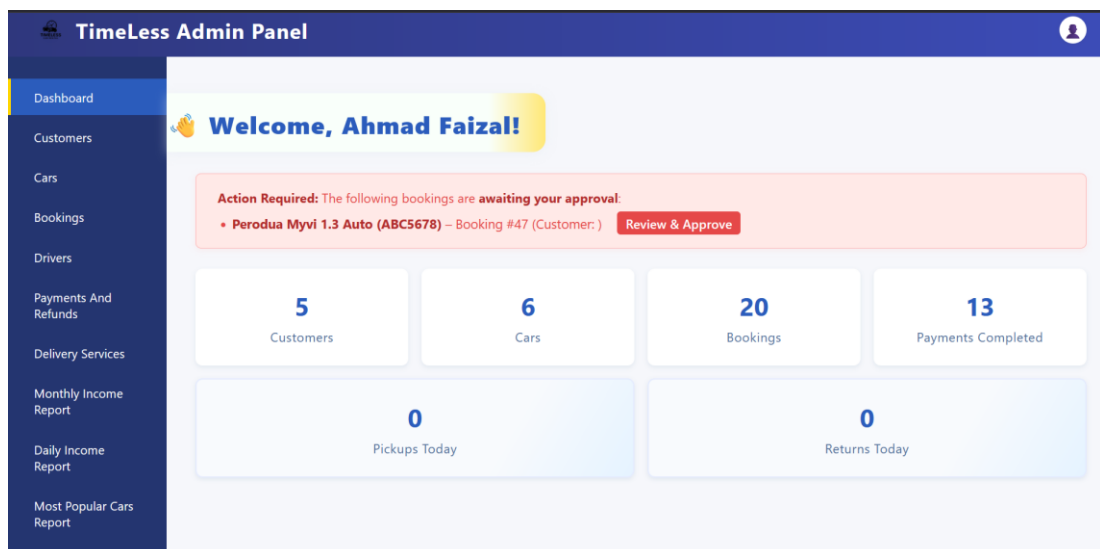


Figure 4.14 Admin Dashboard Page

As shown in the figure 4.13, this page serves as the main dashboard for administrators within the TimeLess system. It welcomes the admin by name and provides a quick

overview of key business metrics, such as the number of customers, cars, total bookings, and payments completed. The dashboard also displays the number of pickups and returns scheduled for the current day.

A prominent notification area highlights any bookings that require immediate approval, allowing the administrator to quickly review and take action on pending requests. The left sidebar provides easy navigation to other sections of the system, including customer management, car inventory, driver details, payments and refunds, and various reports. This layout ensures that administrators can efficiently monitor activity, manage bookings, and access important information, streamlining daily operations within the TimeLess platform.

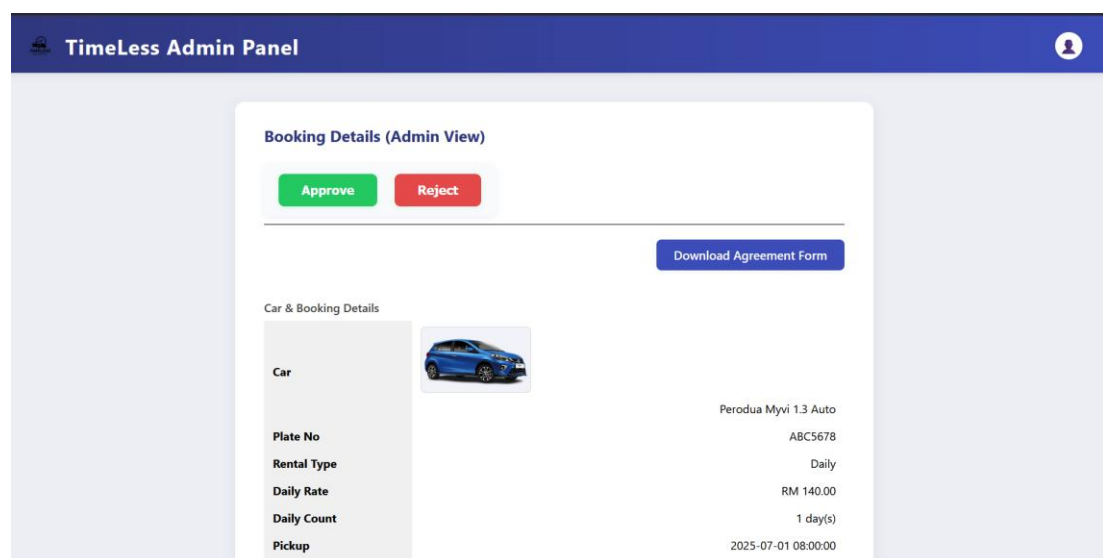


Figure 4.15 Admin Approval Booking Page

As shown in the image, this page presents the detailed view of a car booking for administrators in the TimeLess Admin Panel. It is designed to help admins review and manage individual booking requests efficiently. At the top of the page, there are prominent Approve and Reject buttons, allowing the administrator to quickly take action on the booking. Below these options, there is a button to download the agreement form related to the booking.

TimeLess Admin Panel

Pickup Inspection for Perodua Myvi 1.3 Auto (ABC5678)

Mileage (km): 97000 Fuel Percent (%): 60% Inspection Date: 30/06/2025, 03:04

Inspection Image: Browse... Screensh...4653.png Image Type: Car Front

Inspection Image: Browse... Screensh...4653.png Image Type: Car Back

Inspection Image: Browse... Screensh...4653.png Image Type: Car Left Side

Inspection Image: Browse... Screensh...4653.png Image Type: Car Right Side

Inspection Image: Browse... Screensh...4653.png Image Type: Fuel Image

+ Add More Images

Remarks (optional, add once only):

Submit Inspection

Figure 4.16 Inspection Pickup and Return Page

As shown in the image, this page is used by administrators to conduct and record a pickup inspection for a rental car in the TimeLess Admin Panel. The form allows the admin to enter the car's current mileage (in kilometers). Select the current fuel level as a percentage. Set the inspection date and time. Upload multiple inspection images for different angles or parts of the car (such as Car Front, Car Back, Car Left Side, Car Right Side, and Fuel Image). Each image can be assigned a type to specify what it shows. Add additional images if needed by clicking "+ Add More Images". Provide remarks or comments about the inspection in an optional field.

TimeLess Admin Panel

Dashboard / Customers

Customers

Search name, username, phone, € Search

#	Name	Phone	Email	Username	Address	Age	Edit
1		0199590828	fathehahcomey@gmail.com	fathehaharis13		23	Edit
2	Syamimi Fathehah Binti Karim	0135309008	syamimi@gmail.com	fathehaharis1	Durian Tunggal, Melaka	23	Edit
3	Nur Amiera Badriesya Binti Samad	0175685078	amiera@gmail.com	amieraaa	Durian Tunggal, Melaka	20	Edit
4	amiera khan	0187665678	lmk@gmail.com	littlemisskhan	8-11-6, Jalan Bukit Jalil Indah 1, Taman LTAT, 57000, Kuala Lumpur	20	Edit
5	Nur Fathehah Binti Mohd Aris	0199590828	fathehaharis69@gmail.com	fathehaharis	10, Dataran Tasek Timur 88, Anjung Bercham Indah, 31400 Ipoh, Perak	23	Edit

Figure 4.17 Admin Manage Customer Page

As shown in the image, this page allows administrators to manage customer information within the TimeLess Admin Panel. The page displays a searchable list of all registered customers, showing key details such as name, phone number, email, username, address, and age. Admins can quickly find a specific customer by searching with a name, username, phone, or email using the search bar at the top. Each row in the customer list includes an “Edit” button, enabling the admin to update the selected customer’s information as needed. This setup makes it easy for administrators to efficiently view, locate, and edit customer records, ensuring all customer details are accurate and up-to-date.

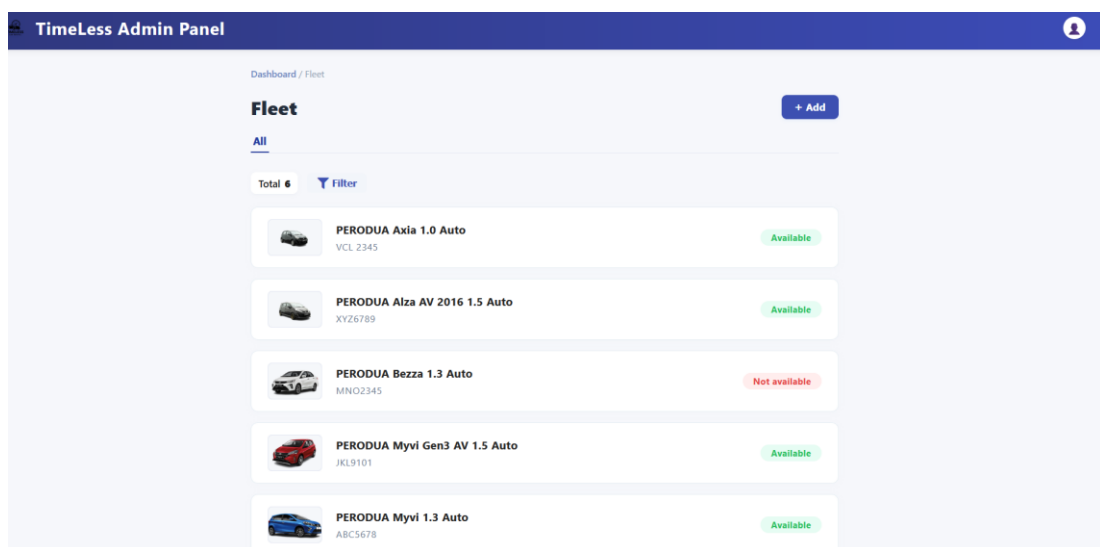


Figure 4.18 Manage Cars Page

As shown in the image, this page allows administrators to manage the fleet of cars within the TimeLess Admin Panel. The Fleet page displays a list of all vehicles currently in the system, along with their status and key details. Admin can add car by enter all the car details and update the car status either available or not available.

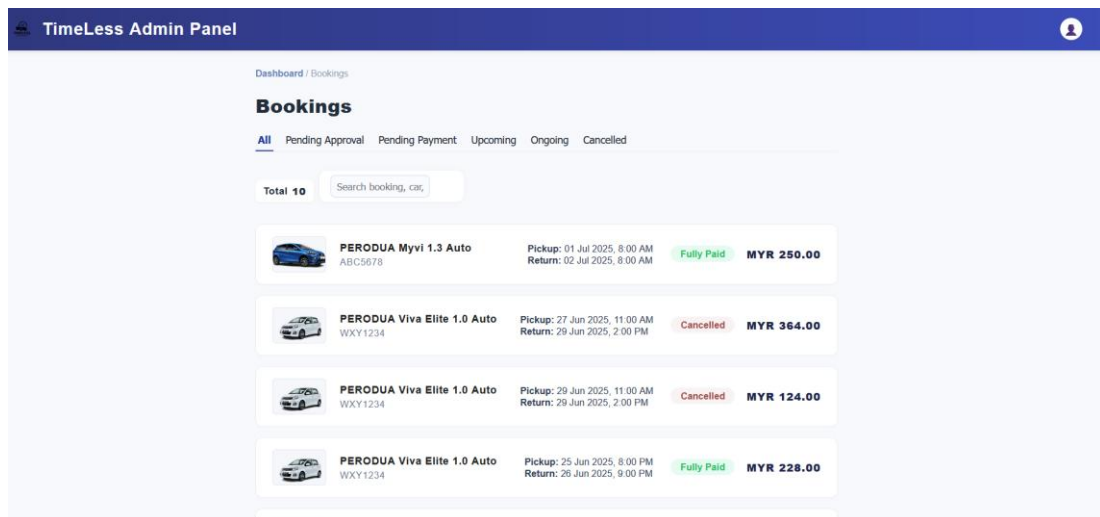


Figure 4.19 Manage Bookings Page

As shown in the image, this page displays the Bookings management section of the TimeLess Admin Panel. Here, administrators can view and manage all car rental bookings in the system. This page gives administrators a clear and organized overview of all bookings, making it easy to monitor booking statuses, track payments, and manage upcoming or ongoing rentals efficiently.

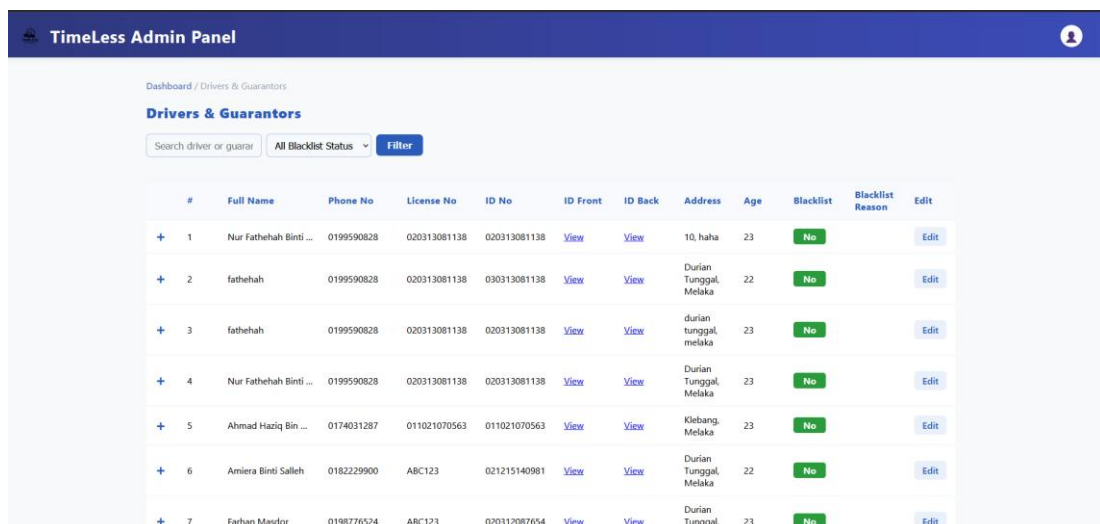


Figure 4.20 Drivers and Guarantor Details Page

This image shows the "Drivers & Guarantors" management page of the TimeLess Admin Panel, where administrators can view, search, and filter a list of drivers and guarantors. The table displays key details for each entry, including full name, phone number, license and ID numbers, links to view ID images, address, age, blacklist status, and options to edit each record. Users can filter the list by blacklist status or

search for specific individuals, and each row provides a quick visual indication of whether the person is blacklisted, along with the ability to edit their information.

#	Amount	Status	Booking Status	Refund Status	Refund Amount	Method	Date	Refund Action
7-02 08:00	MYR 250.00	Paid	Confirmed	-	-	Online Banking - Bank Islam	2025-06-29	-
6-29 14:00	MYR 364.00	Paid	Cancelled	Processed	MYR 264.00	Online Banking - Bank Rakyat	2025-06-24	Processed
6-26 21:00	MYR 228.00	Paid	Confirmed	-	-	Online Banking - Bank Islam	2025-06-25	-
7-03 22:00	MYR 220.00	Paid	Confirmed	-	-	Online Banking - Bank Rakyat	2025-06-24	-
7-10 23:00	MYR 270.00	Paid	Cancelled	Processed	MYR 170.00	Online Banking - Bank Islam	2025-06-23	Processed
6-30 08:00	MYR 240.00	Paid	Cancelled	Processed	MYR 240.00	Online Banking - Bank Islam	2025-06-23	Processed
6-23 14:00	MYR 320.00	Paid	Completed	-	-	Online Banking - Bank Rakyat	2025-06-20	-
6-21 04:00	MYR 240.00	Paid	Completed	-	-	Online Banking - Bank Rakyat	2025-06-21	-
6-20 09:00	MYR 258.00	Paid	Completed	-	-	Credit/Debit Card - 4567	2025-06-19	-
6-26 10:00	MYR 250.00	Paid	Cancelled	Processed	MYR 250.00	Online Banking - Bank Islam	2025-06-18	Processed
7-20 08:00	MYR 540.00	Paid	Cancelled	Processed	MYR 540.00	Online Banking	2025-06-18	Processed
6-21 22:00	MYR 290.00	Paid	Completed	-	-	Credit/Debit Card	2025-06-20	-
6-22 08:00	MYR 230.00	Paid	Completed	-	-	Online Banking	2025-06-20	-
6-20 20:00	MYR 228.00	Paid	Completed	-	-	Online Banking	2025-06-19	-

Figure 4.21 Payment and Refunds Page

This image displays the "Payments & Refunds" section of the TimeLess Admin Panel, where administrators can monitor and manage payment transactions and their associated refund statuses.

Add Delivery Staff

Username*

Full Name*

Password*

Status: Active

[Add Staff](#)

#	Username	Full Name	Status	Action
1	SRina	sabrina binti ahmed	Active	Edit Delete
2	Sfarhan	Mohd Farhan Bin Roslan	Active	Edit Delete

Figure 4.22 Add Delivery Staff Page

Figure 4.21 displays the "Delivery Staff" management page in the TimeLess Admin Panel. At the top, there is a form to add new delivery staff by entering a username, full name, password, and selecting the status (defaulting to Active). Below the form, there is a table listing current delivery staff, showing their username, full name, and status.

For each staff member, there are options to edit or delete their record, enabling easy management of the delivery workforce.

Services & Delivery Assignment									
Search service, custom		Filter							
#	Service Type	Booking ID	Customer	Car	Fee (RM)	Pickup DateTi...	Return DateTi...	Location	Status
1	delivery	47	fathehaharis13	Perodua Myvi 1...	10.00	01/07/2025 08:00	02/07/2025 08:00	SC 1	Pending
2	delivery	41	littlemisskhan	Perodua Alza AV...	10.00	17/07/2025 22:00	18/07/2025 22:00	Sc 1	Out For Delivery
3	pickup_and_r...	30	fathehaharis1	Perodua Myvi G...	30.00	26/06/2025 14:00	27/06/2025 14:00	Scientex1	Pending
4	pickup_and_r...	29	fathehaharis1	Perodua Myvi G...	30.00	20/06/2025 14:00	21/06/2025 14:00	Scientex 2	Delivered
5	pickup_and_r...	28	fathehaharis1	Perodua Axia 1.0...	30.00	25/06/2025 10:00	26/06/2025 10:00	Melaka Sentral	Pending
6	pickup_and_r...	27	fathehaharis1	Perodua Viva Elit...	30.00	19/06/2025 08:00	20/06/2025 09:00	Melaka Sentral	Delivered
7	pickup_and_r...	25	amieraaa	Perodua Bezza 1...	30.00	20/06/2025 22:00	21/06/2025 22:00	Scientex 1	Delivered
8	delivery	24	littlemisskhan	Perodua Axia 1.0...	10.00	21/06/2025 08:00	22/06/2025 08:00	melaka sentral	Delivered

Figure 4.23 Services and Delivery Assignment

Figure 4.22 shows the "Services & Delivery Assignment" page, where delivery or service jobs are tracked. The table lists all assignments with details such as service type (delivery or pickup), booking ID, customer username, car used, fee, pickup and return times, location, and current status (e.g., Pending, Delivered, Out of Delivery). This page helps administrators monitor ongoing and completed assignments, manage workloads, and track delivery staff performance and service fulfillment across different locations and customers.

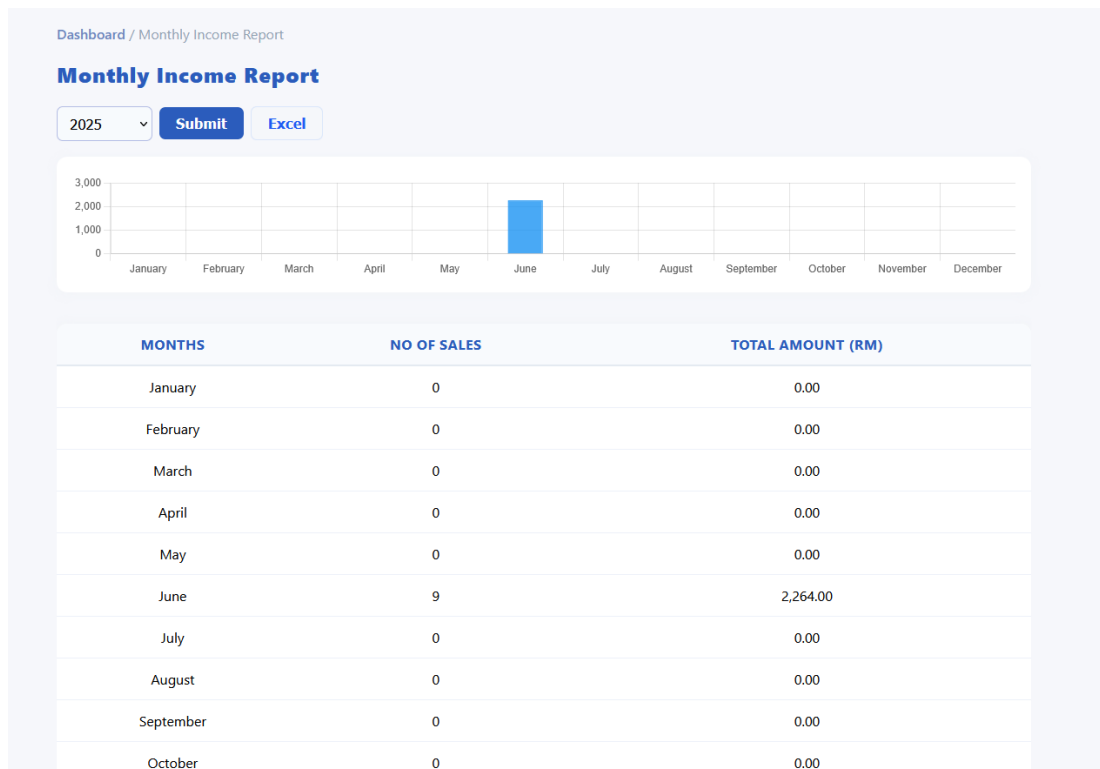


Figure 4.24 Monthly Income Report Page

This image shows the "Monthly Income Report" page from an admin dashboard, summarizing sales and income data for the year. At the top, admin can select the year and either submit the query or export the report to Excel.



Figure 4.25 Most Popular Cars Report Page

This image shows the "Most Popular Cars Report" page in the admin dashboard for the year 2025. At the top, there is a dropdown to select the year, a Submit button to view data for the selected year, and an Excel button to export the report. Below this, a bar chart visually summarizes the number of bookings for each car model, indicating which cars are booked most often.

4.4.3 Delivery Staff Interface

Delivery Staff Portal

TIMELESS
CAR RENTAL

Access your assigned deliveries

Staff Login

Enter your username and password to login.

Username

Password

☐ Remember me

Figure 4.26 Delivery Staff Login

As shown in Figure 4.1, this page provides a login interface specifically for delivery staff of the Timeless Car Rental system. It features a modern, user-friendly design that allows users to sign in using their username and password to access their accounts. The left section displays the company's branding, a promotional tagline, while the right section contains the login form with options such as "Remember me".

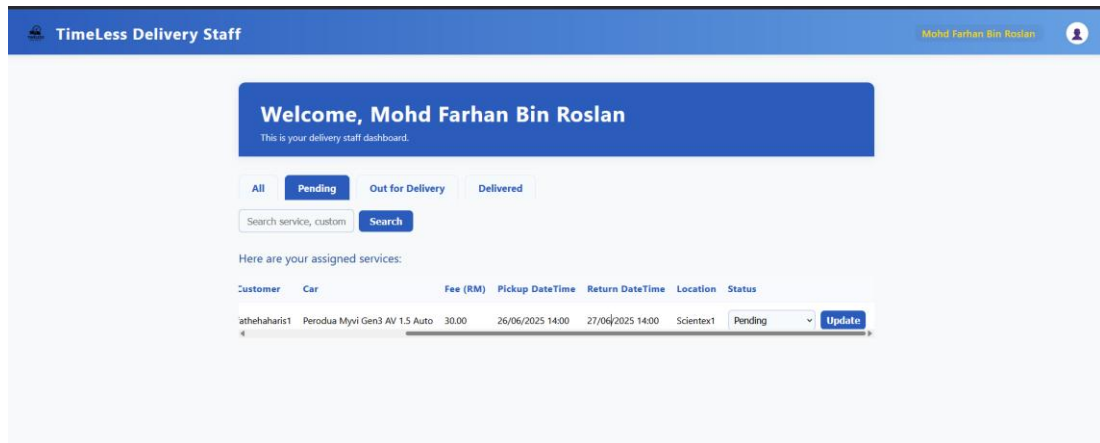


Figure 4.27 Delivery Staff Dashboard Page

This image shows the dashboard for a delivery staff of the TimeLess Delivery Staff system. It welcomes the user and displays a set of tabs to filter delivery assignments by their status (All, Pending, Out for Delivery, Delivered), with the "Pending" tab currently active. There is a search bar to look up services or customers. Each assignment also includes a dropdown menu to update the delivery status and a button to confirm changes, allowing the staff to manage and track their delivery jobs efficiently.

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